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Yamamoto et al.

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(45) **Date of Patent: Aug. 26, 2025**

(54) **GENERAL PRINTING MANAGEMENT DEVICE, CONVEYANCE MANAGEMENT DEVICE, AND PRINTING SYSTEM**

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(72) Inventors: **Hiroki Yamamoto**, Takashima (JP); **Kohei Kashiwagi**, Takashima (JP)

(73) Assignee: **Horizon Inc.**, Takashima (JP)

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B65H 29/22 (2006.01)
B65H 5/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***B65H 29/22*** (2013.01); ***B65H 5/04*** (2013.01); ***B65H 29/00*** (2013.01); ***B65H 31/18*** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC B65H 29/22; B65H 5/04; B65H 29/00; B65H 31/18; B65H 43/00;
(Continued)

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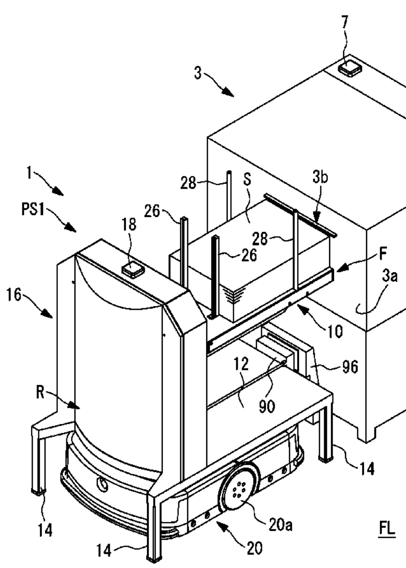
(57) **ABSTRACT**

Personnel saving of a process from a printer to a sheet processing machine is achieved. A general printing management device is connected to be communicable with a conveyance management device that manages a plurality of automatic guided vehicles each conveying a stacker in which paper ejected from a printer is stackable. The general printing management device generates conveyance instruction information for conveying a stacker in which paper ejected from the printer is stacked from a sheet receiving position of the printer to a sheet supply position of a next processing machine, based on job information in which a manufacturing process procedure for manufacturing a printed product is registered, and transmits the conveyance instruction information to the conveyance management device.

12 Claims, 21 Drawing Sheets



(51) **Int. Cl.**
B65H 29/00 (2006.01)
B65H 31/18 (2006.01)
B65H 43/00 (2006.01)

(52) **U.S. Cl.**
CPC ***B65H 43/00*** (2013.01); *B65H 2301/421* (2013.01); *B65H 2301/42252* (2013.01); *B65H 2301/42256* (2013.01); *B65H 2301/54* (2013.01); *B65H 2405/312* (2013.01); *B65H 2511/13* (2013.01); *B65H 2511/18* (2013.01); *B65H 2511/414* (2013.01); *B65H 2511/415* (2013.01); *B65H 2557/25* (2013.01); *B65H 2557/30* (2013.01)

(58) **Field of Classification Search**
CPC B65H 2301/421; B65H 2301/42252; B65H 2301/42256; B65H 2301/54; B65H 2405/312; B65H 2511/13; B65H 2511/18; B65H 2511/414; B65H 2511/415; B65H 2557/25; B65H 2557/30; B25J 9/1664
See application file for complete search history.

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FIG. 1

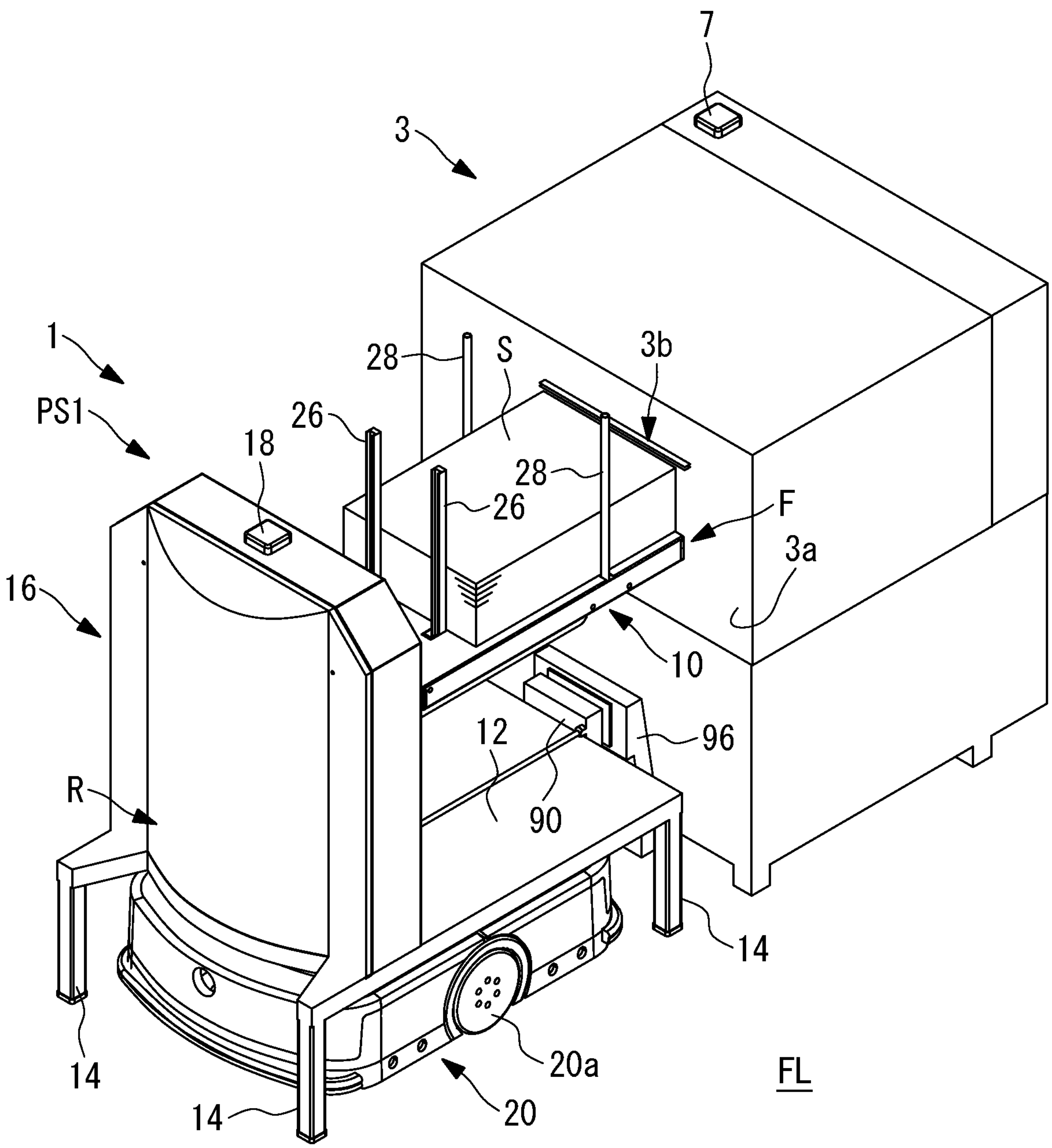


FIG. 2

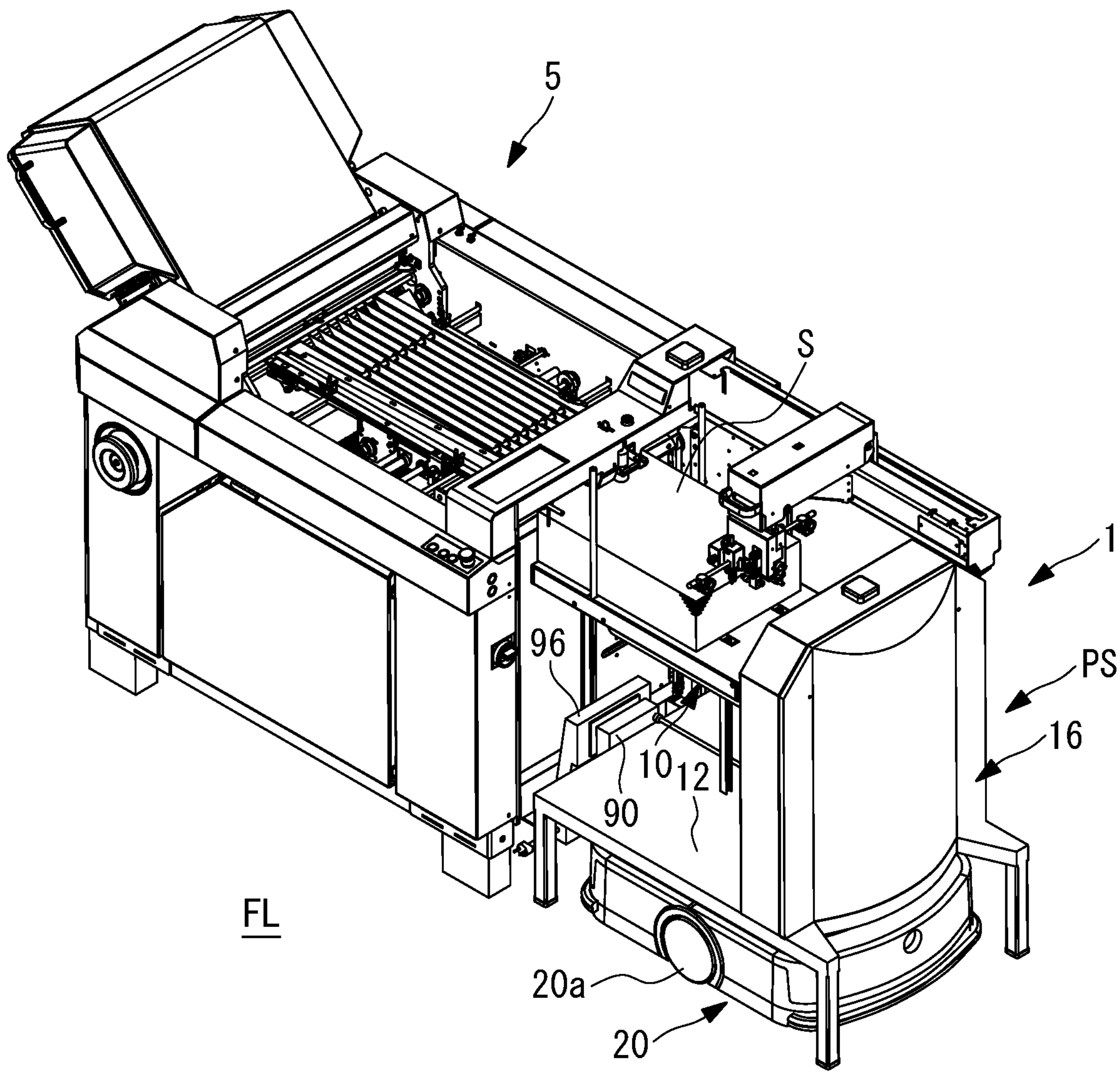


FIG. 3

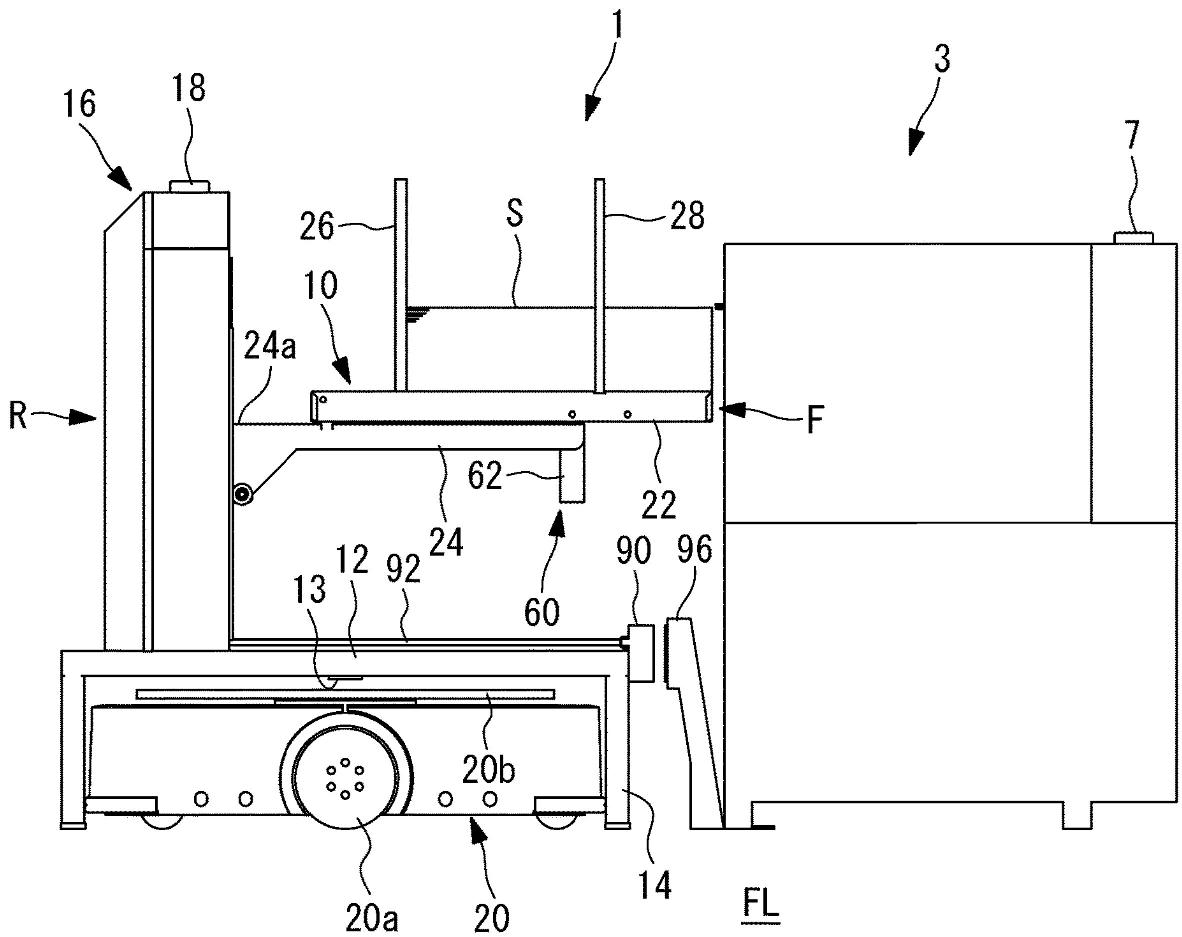
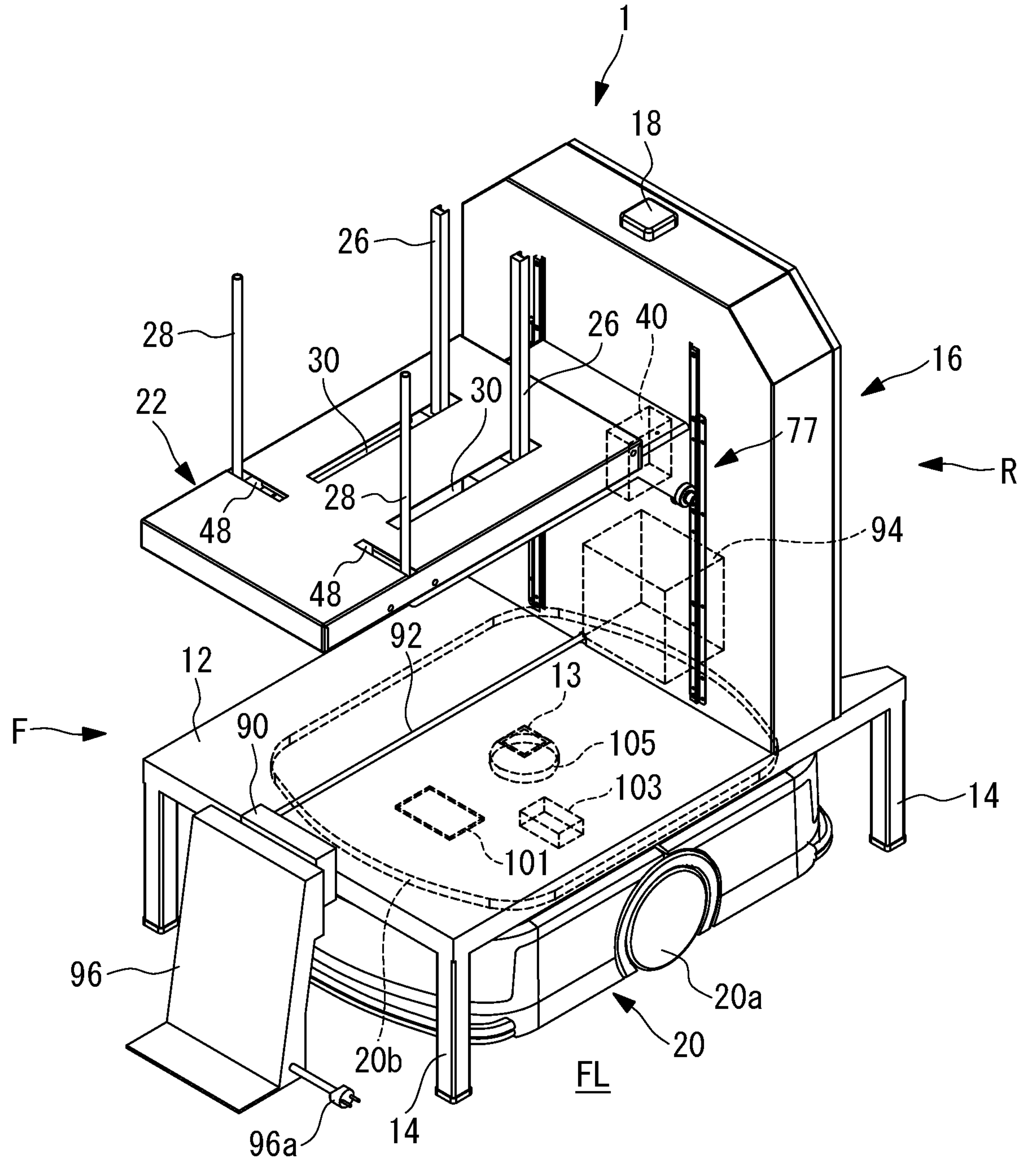


FIG. 4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26
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77
R
48
28
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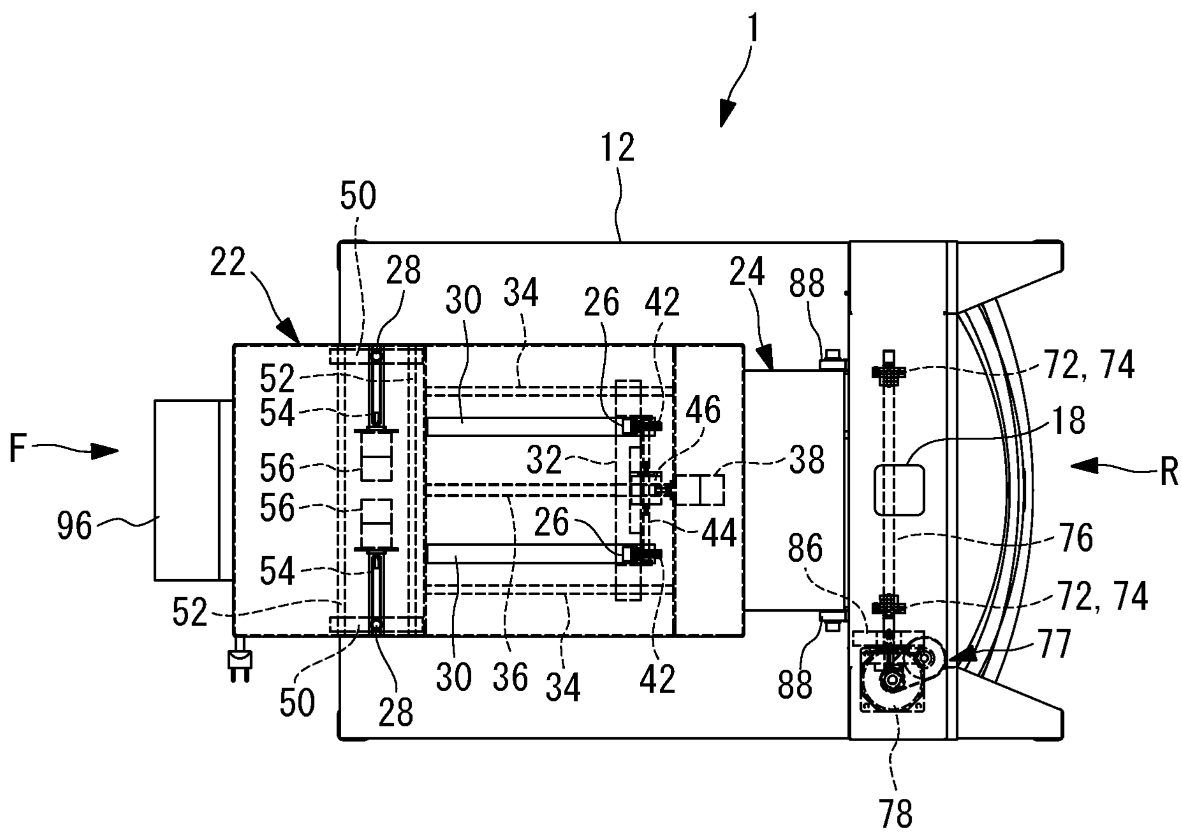
F
90
105
103
14
101
96
20a
20
20b
FL
96a
14

FIG. 5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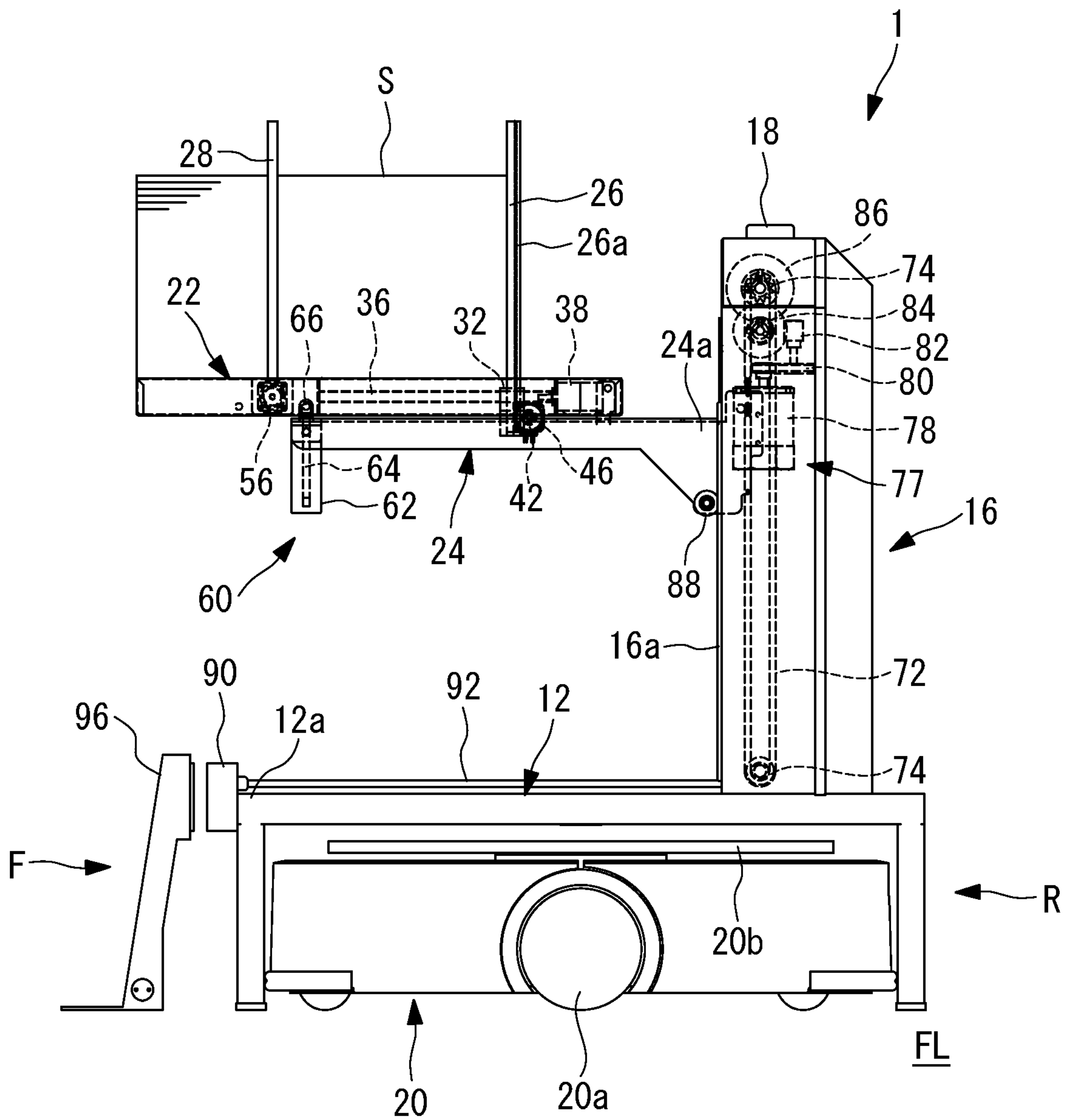
44
26
86
72, 74
18
76
72, 74
77
F
R
96
50
28
30
36
34
42
88
78

FIG. 6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26a
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56
64
42
46
62
24
88
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16a
72
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12
96
12a
74
F
R
20b
FL
20
20a

FIG. 7

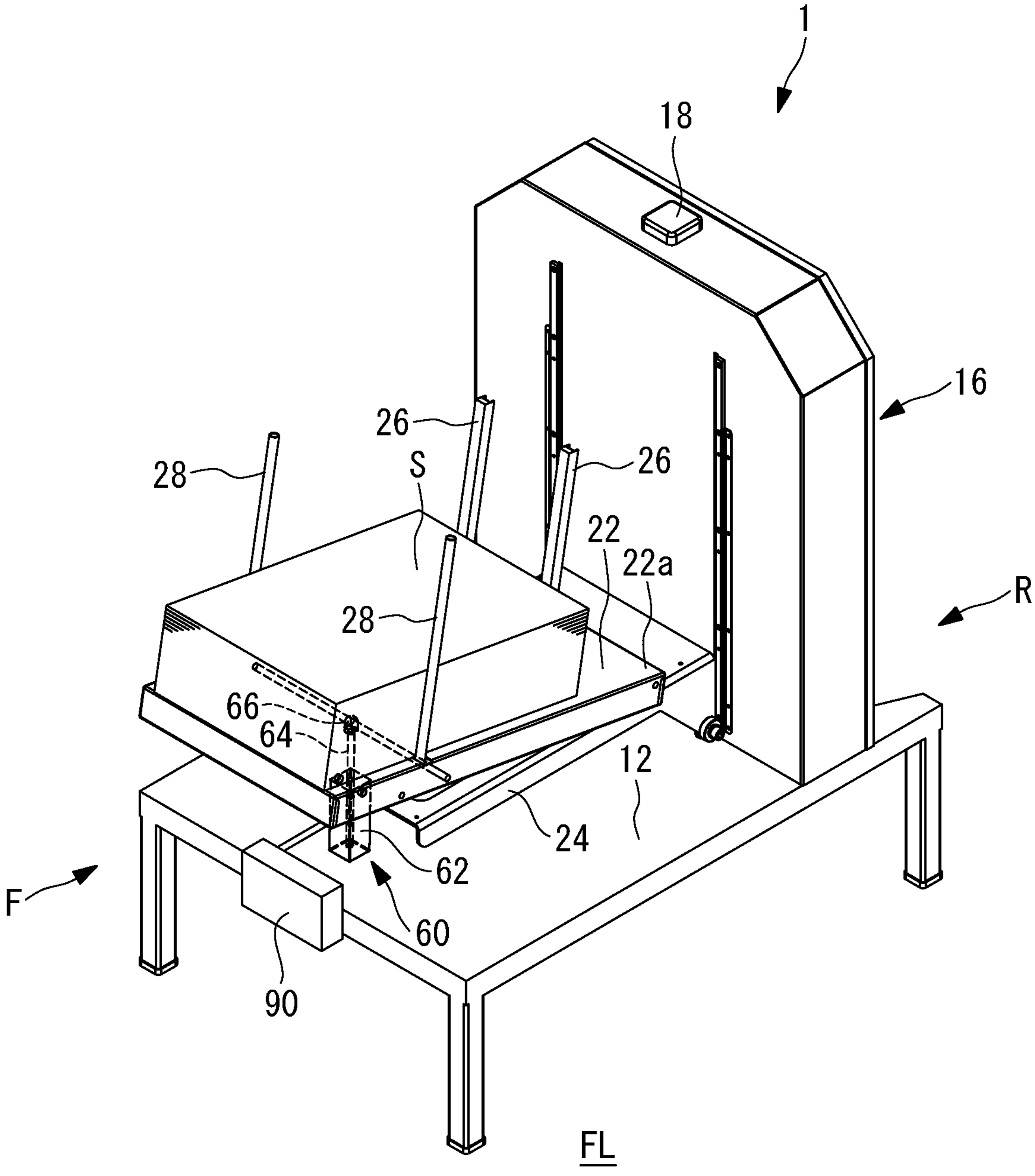


FIG. 8

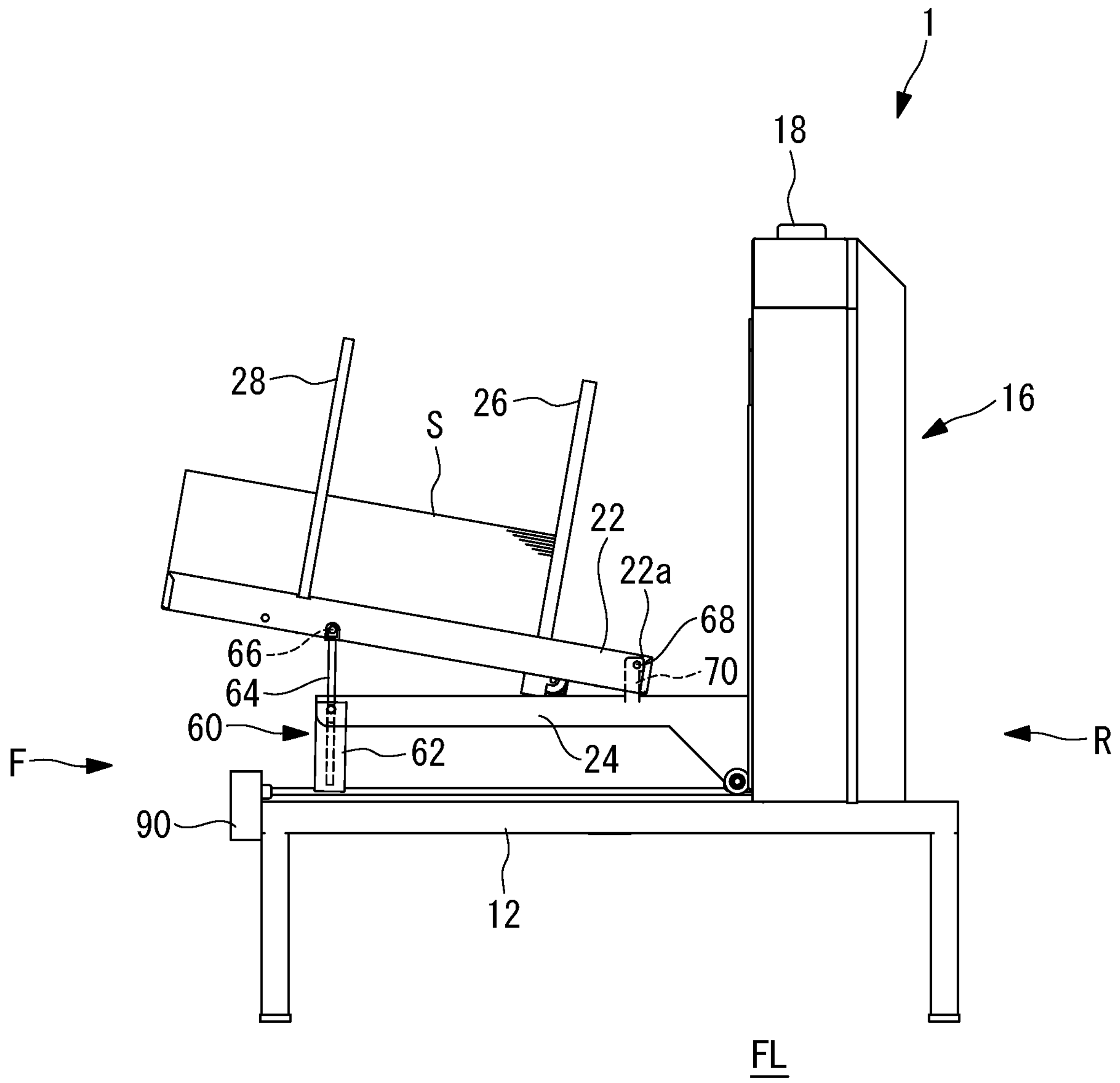
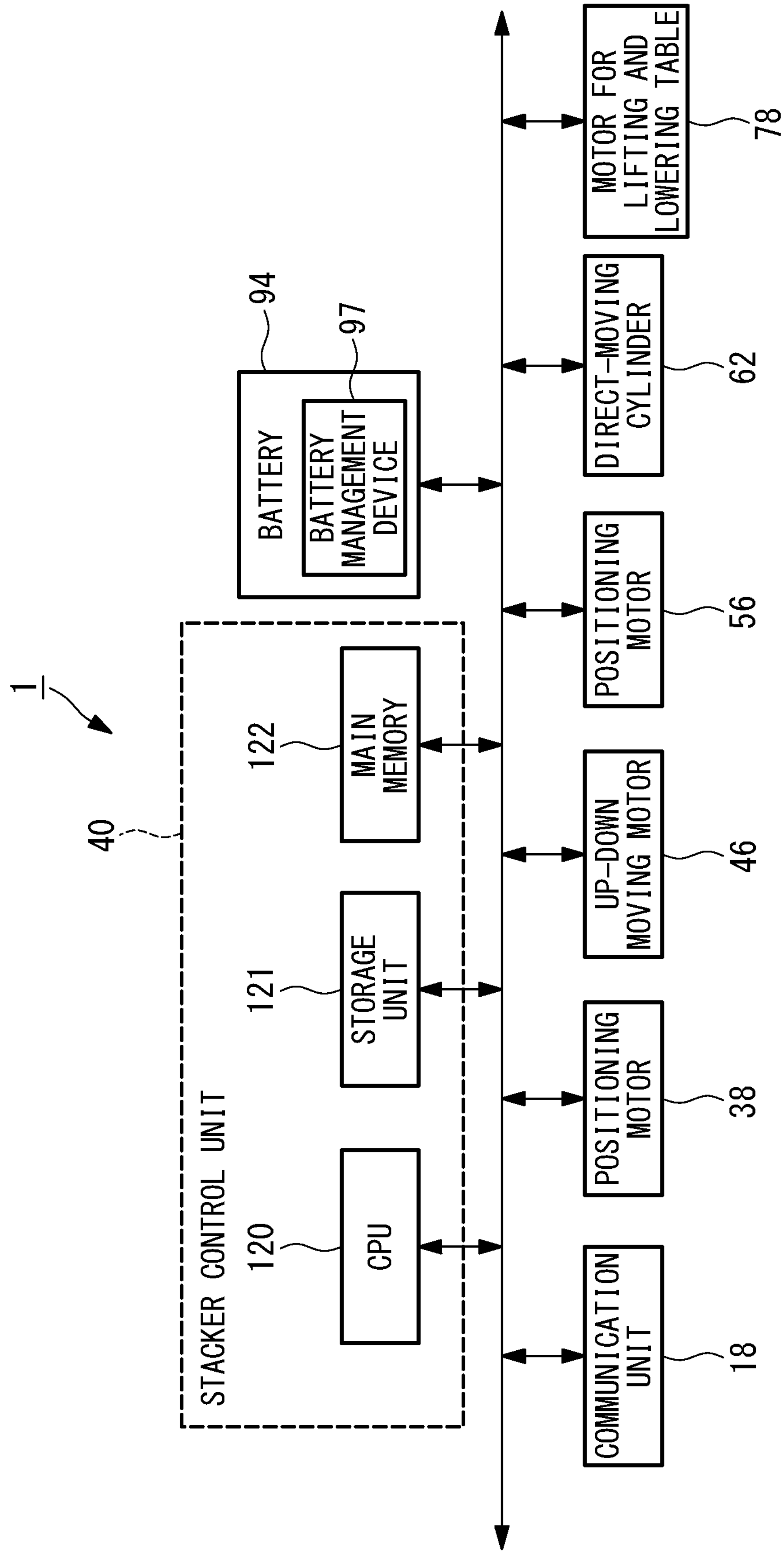


FIG. 9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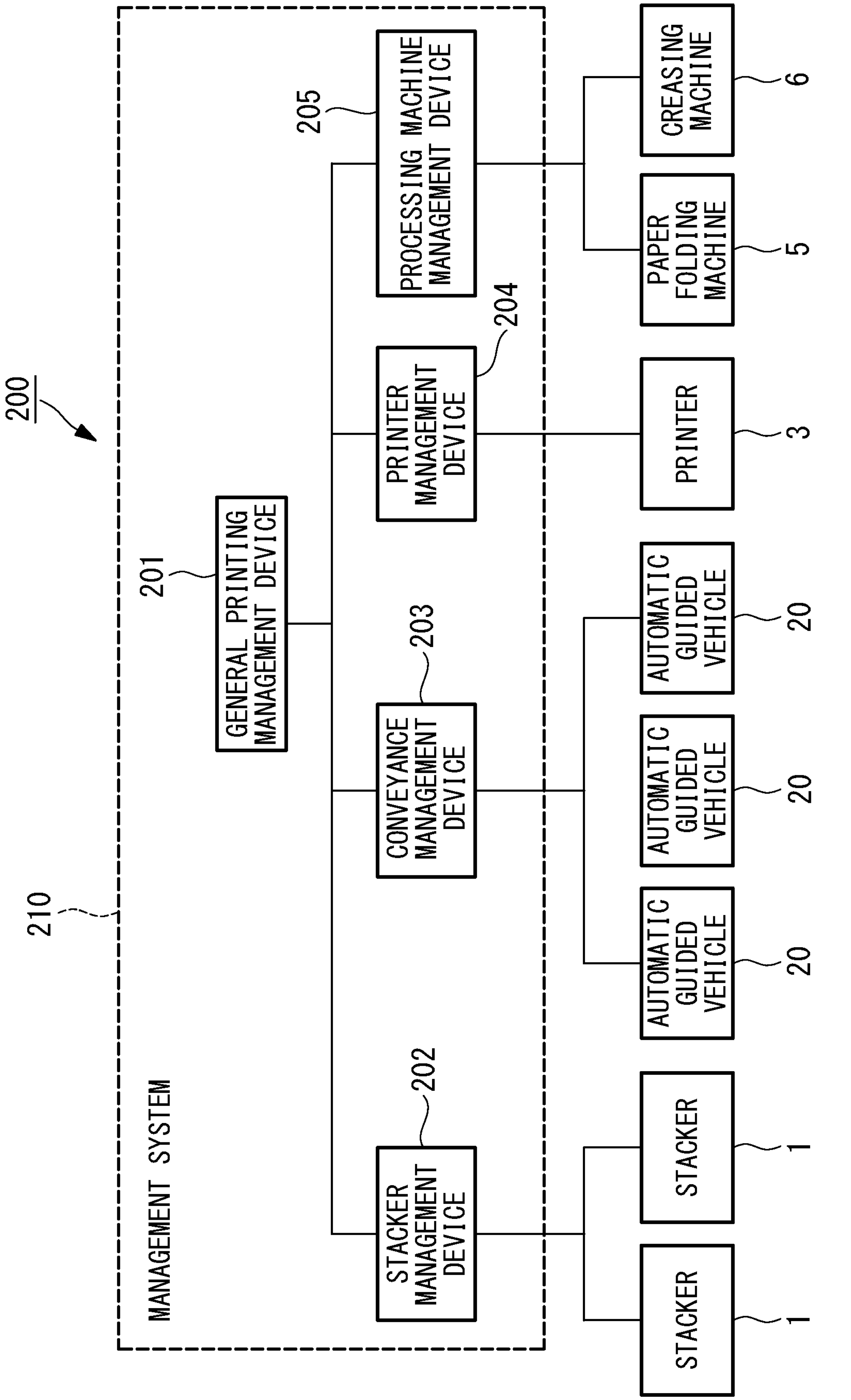
40
STACKER CONTROL UNIT
120
CPU
121
STORAGE UNIT
122
MAIN MEMORY
94
BATTERY
97
BATTERY MANAGEMENT DEVICE
COMMUNICATION UNIT
18
POSITIONING MOTOR
38
UP-DOWN MOVING MOTOR
46
POSITIONING MOTOR
56
DIRECT-MOVING CYLINDER
62
MOTOR FOR LIFTING AND LOWERING TABLE
78

FIG. 10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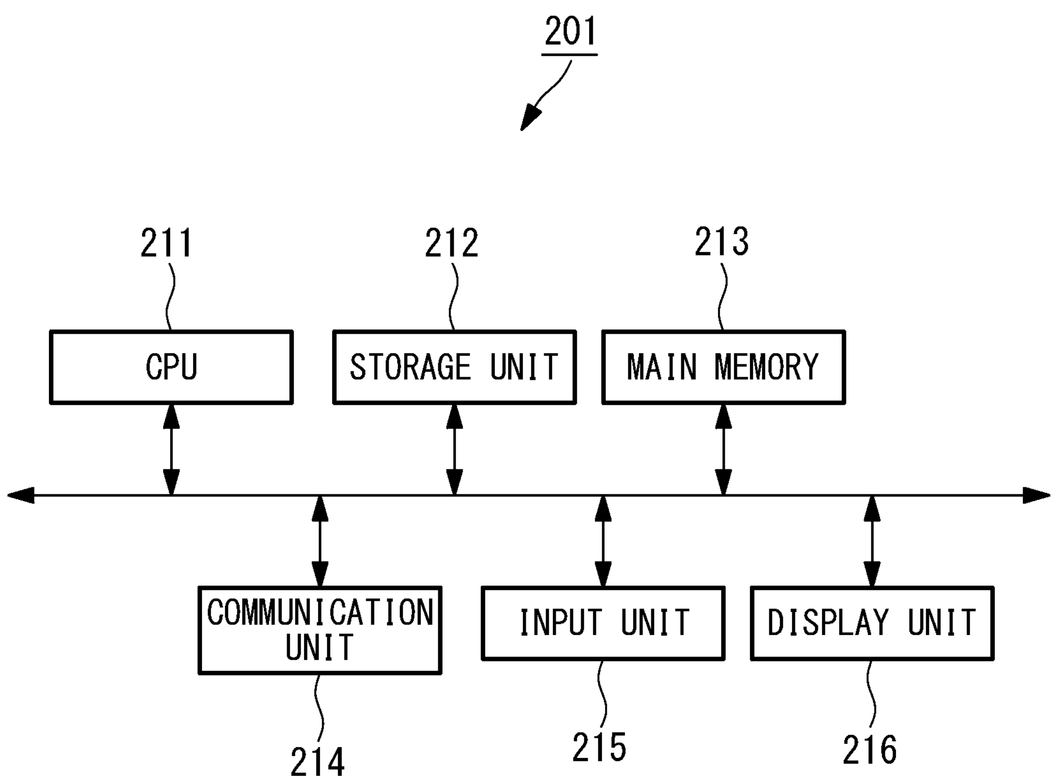
210
MANAGEMENT SYSTEM
201
GENERAL PRINTING MANAGEMENT DEVICE
202
STACKER MANAGEMENT DEVICE
203
CONVEYANCE MANAGEMENT DEVICE
204
PRINTER MANAGEMENT DEVICE
205
PROCESSING MACHINE MANAGEMENT DEVICE
STACKER
1
STACKER
1
AUTOMATIC GUIDED VEHICLE
20
AUTOMATIC GUIDED VEHICLE
20
AUTOMATIC GUIDED VEHICLE
20
PRINTER
3
PAPER FOLDING MACHINE
5
CREASING MACHINE
6

FIG. 11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213
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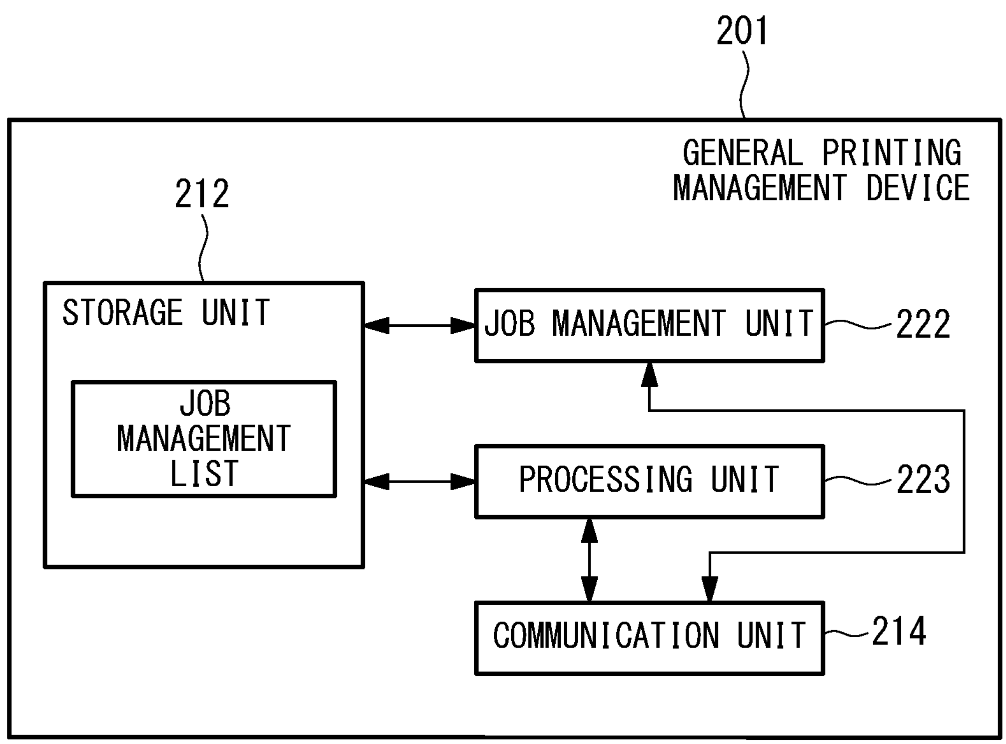
STORAGE UNIT
MAIN MEMORY
COMMUNICATION UNIT
INPUT UNIT
DISPLAY UNIT
214
215
216

FIG. 12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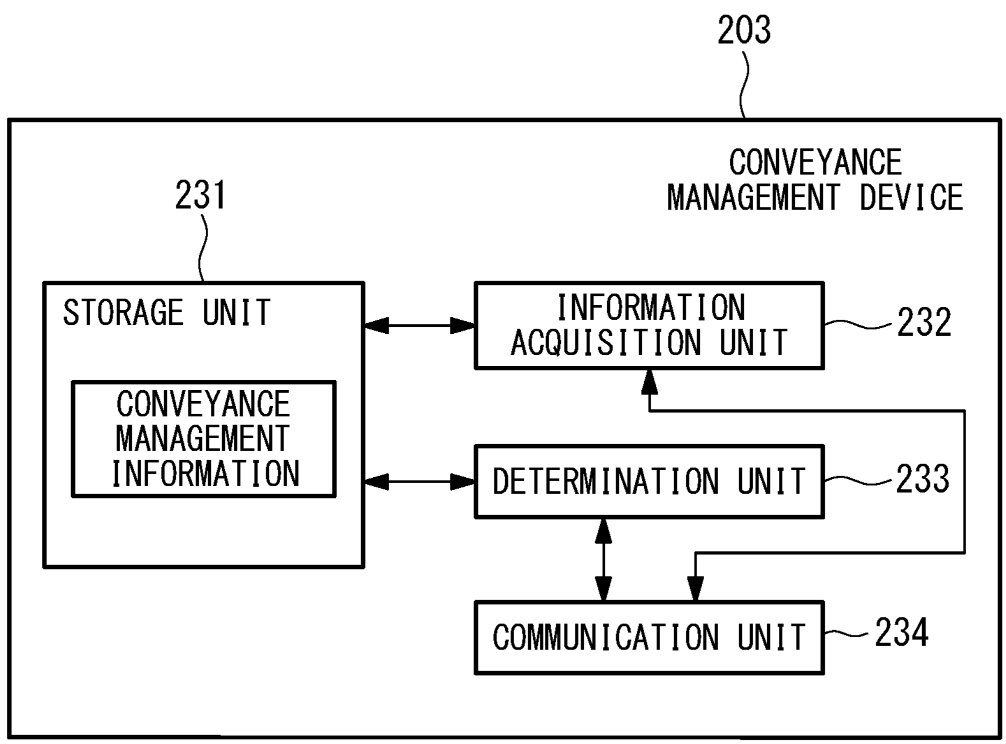
212
STORAGE UNIT
JOB MANAGEMENT LIST
JOB MANAGEMENT UNIT
222
PROCESSING UNIT
223
COMMUNICATION UNIT
214

FIG. 13


203
CONVEYANCE MANAGEMENT DEVICE
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STORAGE UNIT
CONVEYANCE MANAGEMENT INFORMATION
INFORMATION ACQUISITION UNIT
232
DETERMINATION UNIT
233
COMMUNICATION UNIT
234

FIG. 14

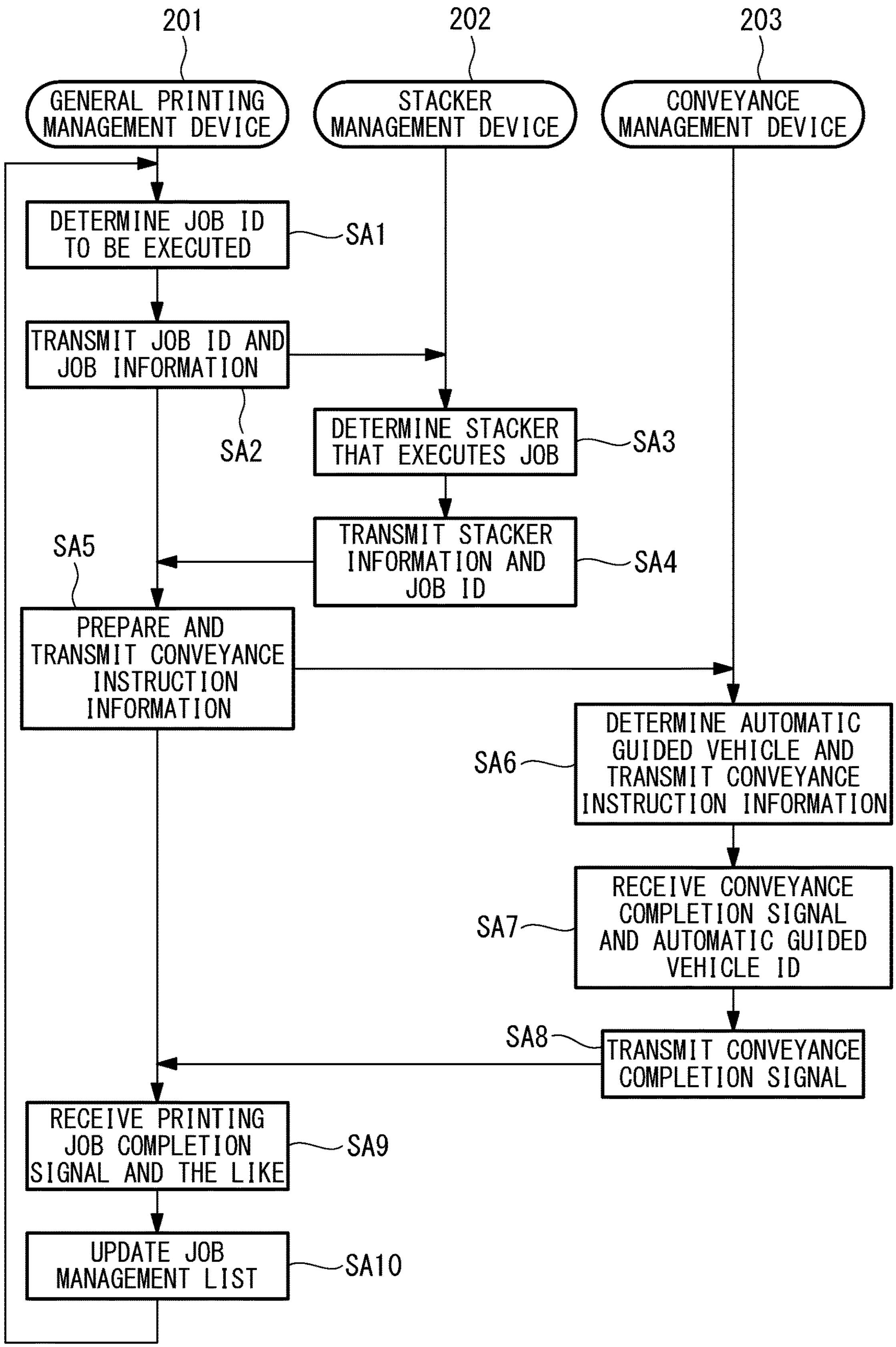

201
202
203
GENERAL PRINTING MANAGEMENT DEVICE
STACKER MANAGEMENT DEVICE
CONVEYANCE MANAGEMENT DEVICE
DETERMINE JOB ID TO BE EXECUTED
SA1
TRANSMIT JOB ID AND JOB INFORMATION
SA2
DETERMINE STACKER THAT EXECUTES JOB
SA3
TRANSMIT STACKER INFORMATION AND JOB ID
SA4
SA5
PREPARE AND TRANSMIT CONVEYANCE INSTRUCTION INFORMATION
DETERMINE AUTOMATIC GUIDED VEHICLE AND TRANSMIT CONVEYANCE INSTRUCTION INFORMATION
SA6
RECEIVE CONVEYANCE COMPLETION SIGNAL AND AUTOMATIC GUIDED VEHICLE ID
SA7
SA8
TRANSMIT CONVEYANCE COMPLETION SIGNAL
RECEIVE PRINTING JOB COMPLETION SIGNAL AND THE LIKE
SA9
UPDATE JOB MANAGEMENT LIST
SA10

FIG. 15

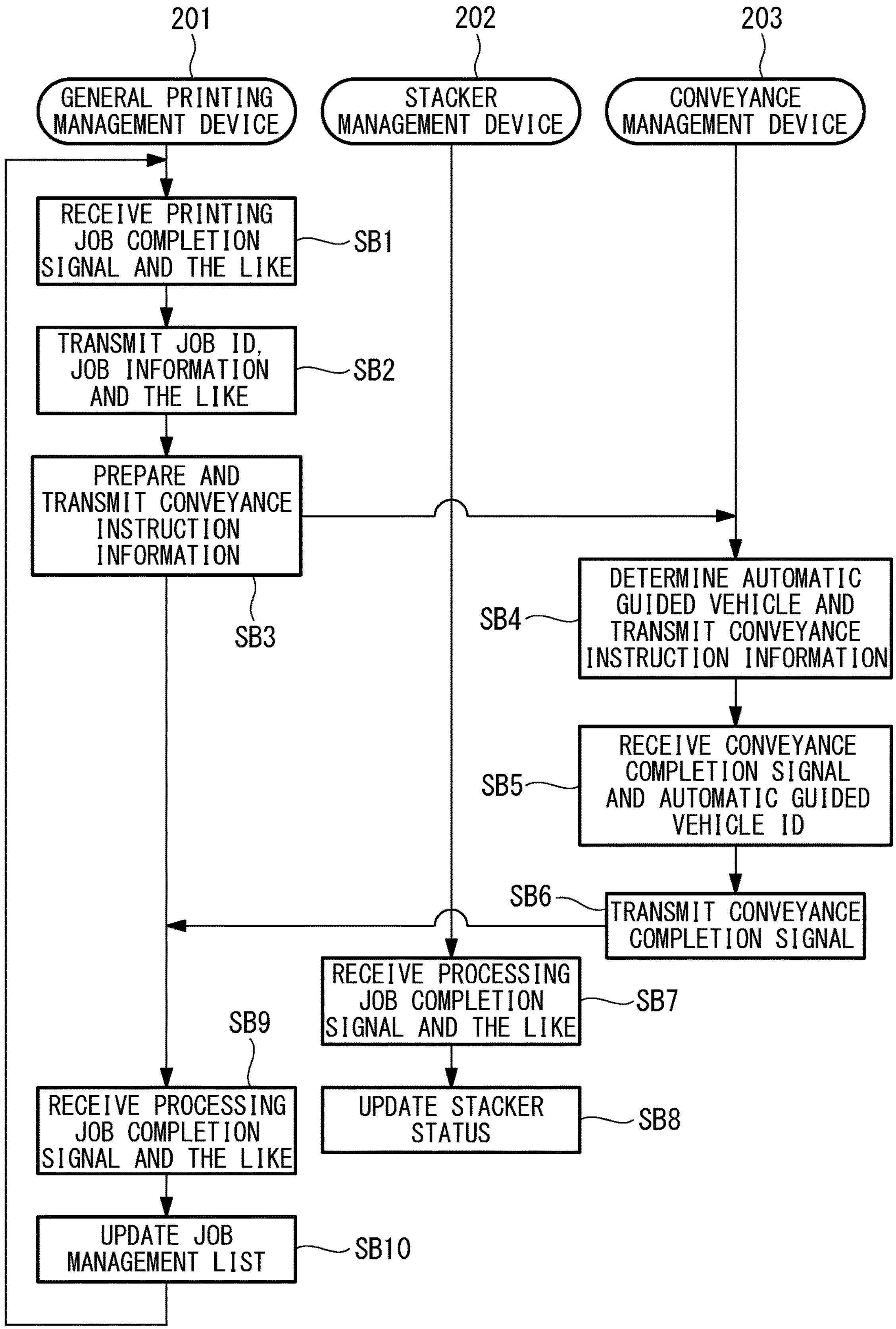

201
202
203
GENERAL PRINTING MANAGEMENT DEVICE
STACKER MANAGEMENT DEVICE
CONVEYANCE MANAGEMENT DEVICE
RECEIVE PRINTING JOB COMPLETION SIGNAL AND THE LIKE
SB1
TRANSMIT JOB ID, JOB INFORMATION AND THE LIKE
SB2
PREPARE AND TRANSMIT CONVEYANCE INSTRUCTION INFORMATION
SB3
DETERMINE AUTOMATIC GUIDED VEHICLE AND TRANSMIT CONVEYANCE INSTRUCTION INFORMATION
SB4
RECEIVE CONVEYANCE COMPLETION SIGNAL AND AUTOMATIC GUIDED VEHICLE ID
SB5
SB6
TRANSMIT CONVEYANCE COMPLETION SIGNAL
RECEIVE PROCESSING JOB COMPLETION SIGNAL AND THE LIKE
SB7
SB9
RECEIVE PROCESSING JOB COMPLETION SIGNAL AND THE LIKE
UPDATE STACKER STATUS
SB8
UPDATE JOB MANAGEMENT LIST
SB10

FIG. 16

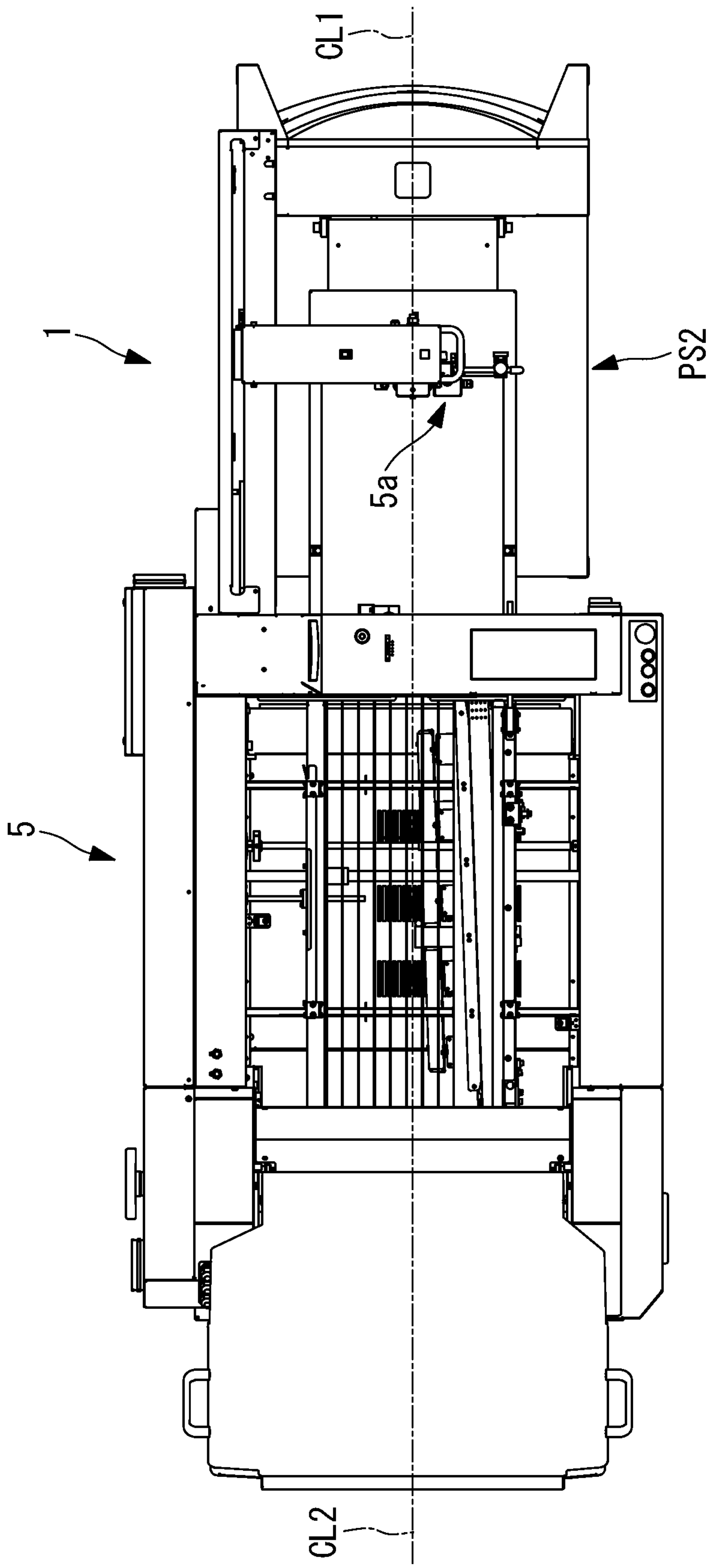


FIG. 17

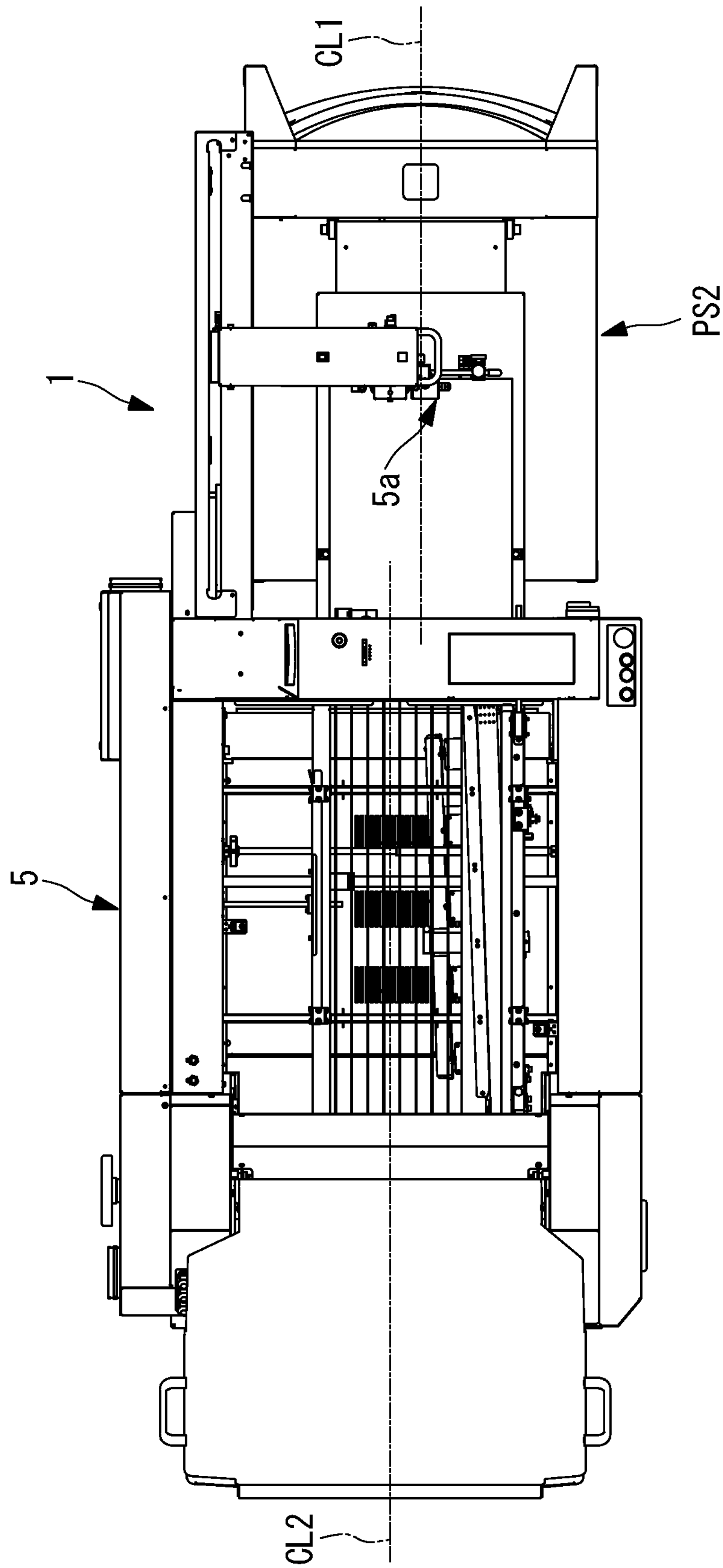


FIG. 18

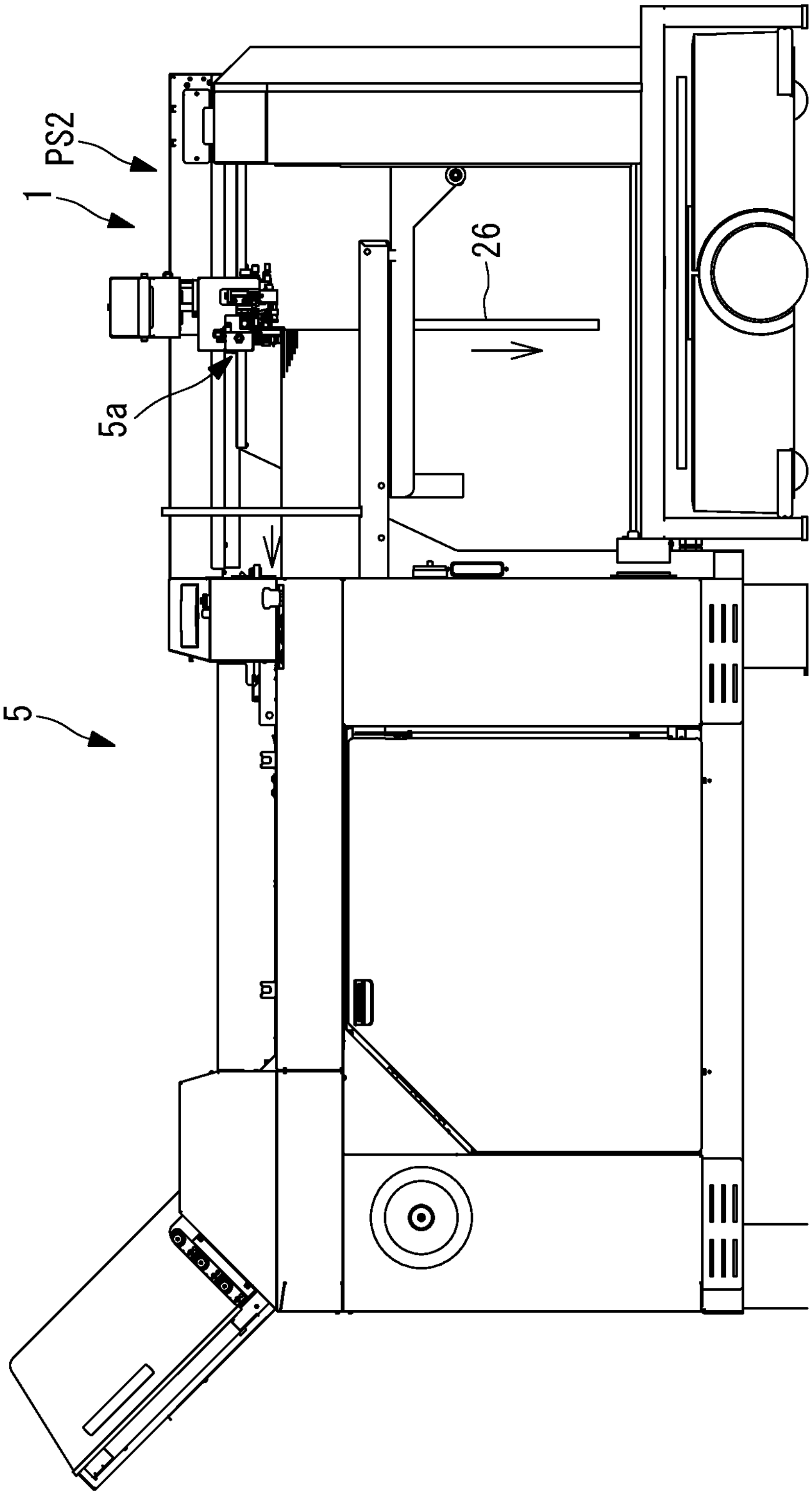


FIG. 19

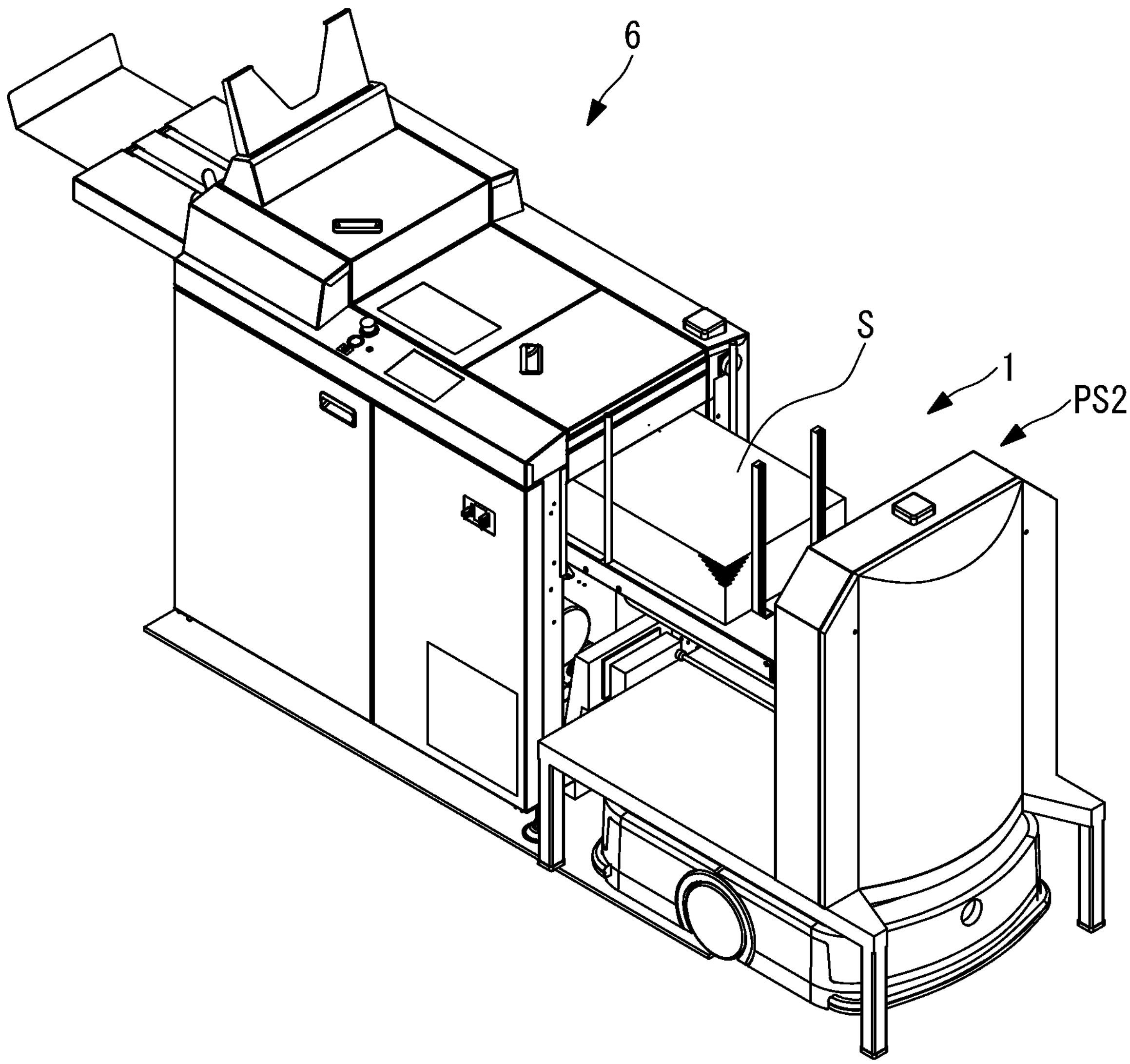


FIG. 20

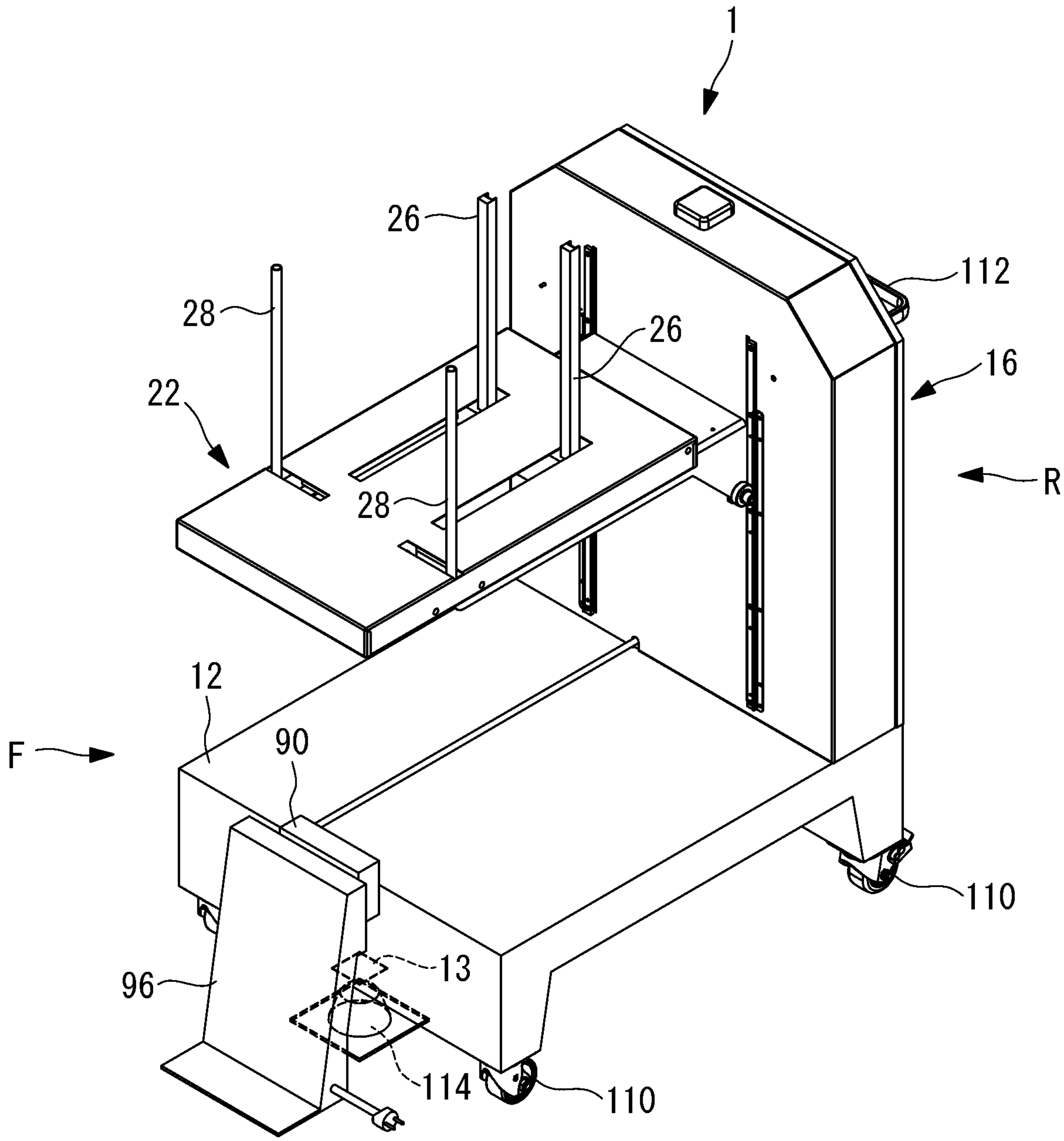
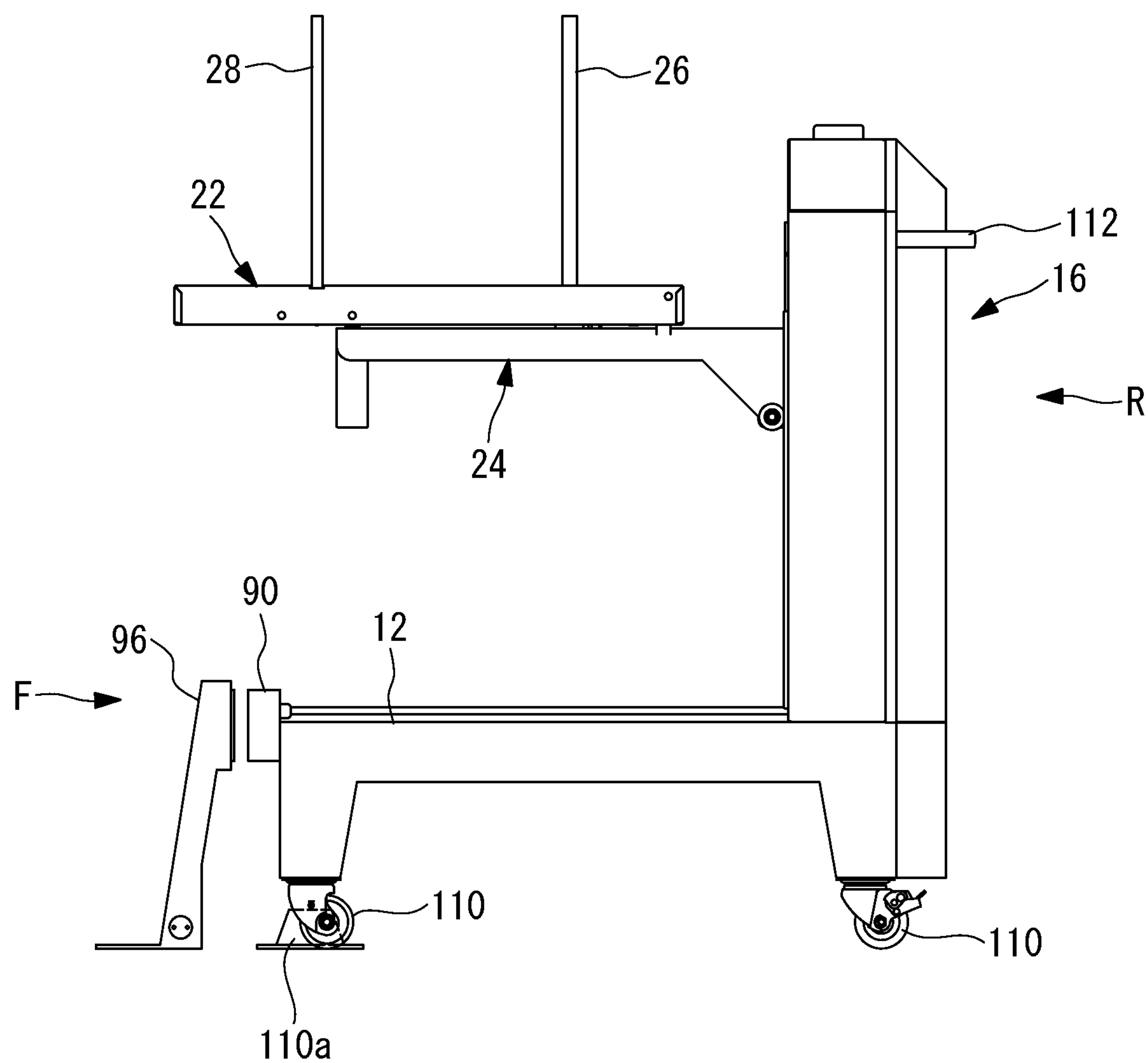


FIG. 21



GENERAL PRINTING MANAGEMENT DEVICE, CONVEYANCE MANAGEMENT DEVICE, AND PRINTING SYSTEM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority to Japanese Patent Application No. 2020-179034 filed Oct. 26, 2020, and Japanese Patent Application No. 2021-144018 filed Sep. 3, 2021, the contents of which are incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to a general printing management device, a conveyance management device, and a printing system.

2. Description of Related Art

Japanese Unexamined Patent Application, Publication No. 2013-52971 discloses a sheet stacking apparatus that stacks paper ejected from a printer on a stacking tray, and conveys the paper to an offline bookbinding machine.

In the sheet stacking apparatus, the stacking tray is lowered depending on a sheet weight, to hold an upper surface of the stacking tray at a fixed position, thereby decreasing burdens on a worker during transshipment work to a paper supply unit.

BRIEF SUMMARY

In a sheet stacking apparatus described in Japanese Unexamined Patent Application, Publication No. 2013-52971, however, movement of a cart or transshipment from the cart to a paper supply unit is performed by a worker, which puts a burden on the worker.

The present disclosure has been made in view of such situation, and an object thereof is to provide a general printing management device, a conveyance management device, and a printing system in which personnel saving can be achieved in a process from a printer to a sheet processing machine.

A first aspect of the present disclosure is a general printing management device connected to be communicable with a conveyance management device that manages a plurality of automatic guided conveying apparatuses each conveying a sheet stacking apparatus in which a sheet ejected from a printer is stackable, the general printing management device including a processor configured to generate conveyance instruction information for conveying the sheet stacking apparatus in which the sheet ejected from the printer is stacked from a sheet receiving position of the printer to a sheet supply position of a next processing machine, based on job information in which a manufacturing process procedure for manufacturing a printed product is registered, and a transmitter configured to transmit the conveyance instruction information to the conveyance management device.

According to the general printing management device, the conveyance instruction information for conveying the sheet stacking apparatus in which the sheet ejected from the printer is stacked from the sheet receiving position of the printer to the sheet supply position of the next processing machine is transmitted to the conveyance management device. Consequently, by use of each automatic guided conveying apparatus, it is possible to automatically move the sheet stacking apparatus from the printer to the processing machine in a next process, and it is possible to achieve personnel saving.

In the general printing management device, the conveyance instruction information may include at least one of a sheet size, a sheet thickness, or the number of sheets to be stacked in the sheet stacking apparatus.

The sheet size, the sheet thickness or the number of the sheets is information concerning a weight of the sheet stacking apparatus, and hence the weight of the sheet stacking apparatus can be estimated based on these pieces of information. Consequently, it is possible to select an appropriate automatic guided vehicle for conveying the sheet stacking apparatus, and it is possible to run the automatic guided conveying apparatus at an appropriate acceleration or speed depending on a stacking weight during the conveying of the sheet stacking apparatus.

In the general printing management device, the conveyance instruction information may include processing machine identification information individually given to the processing machine and offset information of the sheet supply position in the processing machine.

The present inventors have found that personnel saving is obstructed in a case where a type of printer or sheet processing machine varies with a maker, use application or the like of the printer or the machine. For example, in a case of the processing machine, the sheet supply position might vary with the type of processing machine. According to the general printing management device, since the conveyance instruction information includes the processing machine identification information individually given to the processing machine and the offset information of the sheet supply position in the processing machine, it is possible to install the sheet stacking apparatus at an appropriate sheet supply position depending on the type of processing machine. This can achieve stable and smooth sheet supply.

In the general printing management device, the conveyance instruction information may include information concerning a sheet supply direction of the sheet to the processing machine.

Since the conveyance instruction information includes the information concerning the sheet supply direction to the processing machine, it is possible to supply the sheet in an appropriate orientation depending on processing specifications in each processing machine, even in a case where an ejection orientation of the sheet in the printer is different from a sheet supply orientation in the processing machine.

In the general printing management device, the conveyance instruction information may include information concerning an orientation of the sheet to be ejected from the printer to the sheet stacking apparatus.

The conveyance instruction information includes the information concerning the orientation of the sheet to be ejected from the printer to the sheet stacking apparatus, and hence conveyance is performed in consideration of an orientation of the sheet stacked on the sheet stacking apparatus, so that it is possible to prevent sheet collapse during the conveying. For example, to prevent load collapse during accelerating or decelerating, the automatic guided conveying apparatus conveys the sheet stacking apparatus so that a longitudinal direction of the sheet is the same as a traveling direction, and it is therefore possible to prevent the sheet collapse during the conveying.

In the general printing management device, in a case where a plurality of sheet stacking apparatuses are required for execution of one job, the conveyance instruction information may include identification information of the plurality of sheet stacking apparatuses that execute one job, and at least one of an order of the sheet stacking apparatuses in which sheets ejected from the printer are stacked or an order of the sheet stacking apparatuses that supply the sheets to the next processing machine.

According to the general printing management device, even in a case where the number of paper sheets for one job is in excess of a maximum number of the sheets to be stacked in the sheet stacking apparatus, it is possible to smoothly execute the job.

In the general printing management device, the processor may generate, based on the job information, conveyance instruction information for conveying the sheet stacking apparatus in which the sheet is not stacked to the sheet receiving position of the printer.

According to the general printing management device, it is possible to automatically move the sheet stacking apparatus to the printer, and it is possible to achieve further personnel saving.

In the general printing management device, the conveyance instruction information may include identification information individually given to the sheet stacking apparatus.

According to the general printing management device, the conveyance instruction information includes the identification information of the sheet stacking apparatus, and hence, for example, even in a case where a plurality of sheet stacking apparatuses are installed close to one another, the apparatuses are checked based on the identification information. It is therefore possible to prevent incorrect conveyance indicating that the sheet stacking apparatus that is not a conveyance target is incorrectly conveyed.

A second aspect of the present disclosure is a conveyance management device configured to manages a plurality of automatic guided conveying apparatuses each conveying a sheet stacking apparatus in which a sheet ejected from a printer is stackable, and the conveyance management device includes a processor and a memory. The memory including a program that, when executed by the processor, causes the processor to perform operations, the operations including (i) an acquiring at least one of battery information, operating information, or current position information of each automatic guided conveying apparatus, (ii) receiving conveyance instruction information for conveying the sheet stacking apparatus, (iii) determining one automatic guided conveying apparatus based on the information acquired by the acquired information and the conveyance instruction information, and (iv) transmitting the conveyance instruction information to the determined automatic guided conveying apparatus, the conveyance instruction information including at least one of a sheet size, a sheet thickness and the number of sheets to be stacked in the sheet stacking apparatus, identification information of the sheet stacking apparatus, or identification information of the printer or a processing machine that is a conveyance destination of the sheet stacking apparatus.

According to the conveyance management device of the present aspect, the conveyance instruction information includes at least one of the sheet size, the sheet thickness, or the number of the sheets to be stacked in the sheet stacking apparatus, the identification information of the sheet stacking apparatus, or the identification information of the printer or the processing machine that is the conveyance destination of the sheet stacking apparatus. For example, the sheet size, the sheet thickness, and the number of the sheets are information concerning a weight of the sheet stacking apparatus, or hence the weight of the sheet stacking apparatus can be estimated based on these pieces of information. Consequently, it is possible to select an appropriate automatic guided vehicle for conveying the sheet stacking apparatus. Also, in a case where the identification information of the printer or the processing machine is included, concern that the sheet stacking apparatus is conveyed to an incorrect conveyance destination can be reduced by checking equipment based on the identification information.

In the conveyance management device, the conveyance instruction information may include information concerning a running route from a conveyance source to a conveyance destination of the sheet stacking apparatus and information of a particularity on the running route, and the particularity may include at least one of floor slope information, floor step information, temperature, humidity, air conditioning air volume, or air conditioning wind direction.

According to the conveyance management device, the information concerning the running route from the conveyance source to the conveyance destination of the sheet stacking apparatus and the information of the particularity on the running route are transmitted to the automatic guided conveying apparatus. Consequently, the automatic guided conveying apparatus performs adjustment of a speed or an acceleration, reconstruction of the running route, change of an orientation of a stacker relative to the traveling direction or the like depending on the information of the particularity, so that it is possible to inhibit sheet load collapse during the moving.

A third aspect of the present disclosure is a printing system including the general printing management device, and the conveyance management device.

A fourth aspect of the present disclosure is a method of executing by a computer. The method including generating conveyance instruction information for conveying a sheet stacking apparatus in which a sheet ejected from a printer is stacked to a sheet supply position of a next processing machine, based on job information in which a manufacturing process procedure for manufacturing a printed product is registered, and transmitting the conveyance instruction information to a conveyance management device configured to manage a plurality of automatic guided conveying apparatuses each conveying the sheet stacking apparatus.

A fifth aspect of the present disclosure is a non-transitory computer readable storage medium storing a computer program that, when executed by a processor, causes the processor to (i) generate conveyance instruction information for conveying a sheet stacking apparatus in which a sheet ejected from a printer is stacked to a sheet supply position of a next processing machine, based on job information in which a manufacturing process procedure for manufacturing a printed product is registered, and (ii) transmit the conveyance instruction information to a conveyance management device configured to manage a plurality of automatic guided conveying apparatuses each conveying the sheet stacking apparatus.

Advantageous Effects

Personnel saving of a process from the printer to the sheet processing machine can be achieved.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. **1** is a perspective view showing a state where a stacker according to an embodiment of the present disclosure is disposed at a receiving position to a printer.

FIG. **2** is a perspective view showing a state where the stacker is disposed at a paper supply position to a paper folding machine.

FIG. **3** is a side view showing the stacker and printer of FIG. **1**.

FIG. **4** is a perspective view showing the stacker.

FIG. **5** is a plan view showing the stacker of FIG. **4**.

FIG. **6** is a side view showing the stacker of FIG. **4**.

FIG. **7** is a perspective view showing a state where a stacking shelf of the stacker of FIG. **4** is tilted.

FIG. **8** is a side view of FIG. **7**.

FIG. **9** is a block diagram showing an example of a hardware configuration of the stacker according to an embodiment of the present disclosure.

FIG. **10** is a schematic diagram showing an example of an entire configuration of a printing system according to an embodiment of the present disclosure.

FIG. **11** is a block diagram showing an example of a hardware configuration of a general printing management device according to an embodiment of the present disclosure.

FIG. **12** is a functional block diagram showing an example of a function included in the general printing management device according to an embodiment of the present disclosure.

FIG. **13** is a functional block diagram showing an example of a function included in a conveyance management device according to an embodiment of the present disclosure.

FIG. **14** is a flowchart mainly showing an example of a procedure of processing to be executed by the general printing management device, a stacker management device, and a conveyance management device in printed product manufacturing management processing of a management system concerning a printing process according to an embodiment of the present disclosure.

FIG. **15** is a flowchart mainly showing an example of a procedure of processing to be executed by the general printing management device, the stacker management device, and the conveyance management device in the printed product manufacturing management processing of the management system concerning a processing process according to an embodiment of the present disclosure.

FIG. **16** is a plan view showing a state where a center line of the paper folding machine coincides with a center line of the stacker at the paper supply position.

FIG. **17** is a plan view showing a state where the center line of the stacker is offset from the center line of the paper folding machine at the paper supply position.

FIG. **18** is a side view showing a state where a stopper of the stacker is lowered at the paper supply position.

FIG. **19** is a perspective view showing a state where a stacker is disposed at a paper supply position to a creasing machine according to Modification 1.

FIG. **20** is a perspective view showing a stacker according to Modification 2.

FIG. **21** is a side view of the stacker of FIG. **20**.

DETAILED DESCRIPTION

Hereinafter, description will be made as to an embodiment including a general printing management device, a conveyance management device, and a printing system according to the present disclosure with reference to the drawings.

FIG. **1** shows a state where a stacker (sheet stacking apparatus) **1** included in a printing system **200** (see FIG. **10**) according to the present embodiment is disposed at a receiving position PS**1** for receiving paper (sheet) S from a printer **3**.

In the stacker **1**, a predetermined number of sheets of paper S printed in the printer **3** are stacked, and then moved to a paper supply position (supply position) PS**2** for supplying paper to such a paper folding machine (processing machine) **5** as shown in FIG. **2**.

The printer **3** is, for example, a digital printer as shown in FIG. **1**, and receives, in a communication unit **7**, job information from a printer management device **204** (see FIG. **10**), to perform printing of the paper S based on the job information. Note that the job information will be described later in detail.

In a back surface **3***a* of the printer **3**, a paper ejection port **3***b* through which the printed paper (sheet) S is ejected out from the printer **3** is formed. The printer **3** performs the printing of the paper S, and ejects the paper S from the paper ejection port **3***b* toward a shelf unit **10** of the stacker **1**. Also, the printer **3** counts printed paper, and transmits a count number to the printer management device **204**. Further, in a case where the count number reaches the number of sheets to be printed that is included in the job information, a job completion signal and the job ID that is job identification information are transmitted to the printer management device **204**.

As shown in FIG. **3**, the stacker **1** has a rectangular shape in planar view and includes a base **12**. Leg parts **14** are fixed to four corners of the base **12**, respectively. Each leg part **14** is vertically disposed on a floor FL, to support a weight of the stacker **1**. A dimension of each leg part **14** in an up-down direction is a length to such an extent that an automatic guided vehicle **20** of a low floor type can be stored under the base **12**. The automatic guided vehicle **20** lifts the base **12** from below, to convey the stacker **1**. Therefore, the stacker **1** does not include a running device that runs by itself. The automatic guided vehicle **20** includes a wheel **20***a*, and runs along a predetermined route in accordance with an instruction from an after-mentioned conveyance management device **203** (see FIG. **10**).

Stacker ID (identification information) **13** is fixed to a lower surface of the base **12**. In the stacker ID **13**, unique identification information by which the stacker **1** can be identified is recorded or printed. As the stacker ID **13**, an IC chip, a two-dimensional barcode or the like may be used.

On a rear R side of the base **12**, a main body **16** is disposed vertically upward from the base **12**. The main body **16** supports one end of the shelf unit **10** in a cantilever state. A communication unit **18** is disposed on an upper part of the main body **16**.

As shown in FIG. **3**, the shelf unit **10** of the stacker **1** includes a stacking shelf **22** on which the paper S is directly stacked, and a lifting and lowering table **24** located below the stacking shelf **22**. The stacking shelf **22** is a rectangular plate-shaped body in planar view. The stacking shelf **22** includes a stopper **26** and a paper width guide **28**.

The stopper **26** is a rod-shaped body disposed vertically upward from the stacking shelf **22**, and is disposed on the rear R side of the stacking shelf **22**. For example, two stoppers **26** are arranged in a width direction of the stacking shelf **22** as shown in FIG. **1**. Note that the width direction of the stacking shelf **22** is a direction orthogonal to a longitudinal direction of the stacking shelf **22** that is a direction connecting a front F and the rear R. A tip of the paper S ejected from the printer **3** abuts on the stopper **26**, and the paper S is accordingly positioned in an ejection direction.

As shown in FIG. **4**, a lower end side of each stopper **26** is inserted into each of stopper running grooves **30** formed in the stacking shelf **22**. The stopper running groove **30** is formed linearly along the longitudinal direction of the stacking shelf **22**. Each stopper **26** reciprocally moves along the stopper running groove **30**.

As shown in FIG. **5** and FIG. **6**, a lower end of the stopper **26** is fixed to a bracket **32** extending in the width direction of the stacking shelf **22**. Slide guide shafts **34** are inserted into opposite ends of the bracket **32** in the width direction, respectively. Each slide guide shaft **34** is fixed to a stacking shelf **22** side, and extends in the longitudinal direction of the stacking shelf **22**. The bracket **32** is guided along the slide guide shafts **34** to reciprocally move.

The bracket **32** includes a feed screw **36** fixed to a center in the width direction. The feed screw **36** is rotated about an axis by a positioning motor **38** fixed to the rear R side of the stacking shelf **22**. The positioning motor **38** is forward and reverse rotatable in accordance with an instruction of a stacker control unit **40** (see FIG. **4**). The positioning motor **38** rotates the feed screw **36**, to position, in the longitudinal direction, each stopper **26** fixed to the bracket **32**.

As shown in FIG. **6**, an upper and lower rack **26***a* is disposed in an up-down direction on one side of the rear R side of each stopper **26**. A pinion gear **42** disposed in the stacking shelf **22** meshes with each upper and lower rack **26***a*. Each pinion gear **42** is connected to an up-down moving motor **46** via a rotary shaft **44** (see FIG. **5**). The pinion gear **42** is rotated forward and reverse by the up-down moving motor **46** via the rotary shaft **44**, and each stopper **26** including the upper and lower rack **26***a* accordingly moves in the up-down direction. The up-down moving motor **46** is controlled by the stacker control unit **40** (see FIG. **4**).

As shown in FIG. **3**, the paper width guide **28** is a rod-shaped body disposed vertically upward from the stacking shelf **22**, and is disposed on a front F side of the stopper **26**. As shown in FIG. **1**, two paper width guides **28** are arranged to be located on opposite sides in the width direction of the paper S.

As shown in FIG. **4**, a lower end side of each paper width guide **28** is inserted into each of paper width guide running grooves **48** formed in the stacking shelf **22**. Each paper width guide running groove **48** is formed linearly along the width direction of the stacking shelf **22**. Each paper width guide **28** reciprocally moves along the paper width guide running groove **48**.

As shown in FIG. **5**, lower ends of the respective paper width guides **28** are fixed to brackets **50** extending in the longitudinal direction of the stacking shelf **22**, respectively. Slide guide shafts **52** are inserted into opposite ends of each bracket **50** in the width direction. Each of the slide guide shafts **52** is fixed to the stacking shelf **22** side, and extends in the width direction of the stacking shelf **22**. The bracket **50** is guided by each slide guide shaft **52** to reciprocally move.

Each of feed screws **54** is attached to a center of each bracket **50** in the longitudinal direction. The feed screws **54** are rotated about axes by positioning motors **56** fixed to the stacking shelf **22**. Each positioning motor **56** is forward and reverse rotatable in accordance with the instruction of the stacker control unit **40** (see FIG. **4**). The positioning motor **56** rotates the feed screw **54**, so that each paper width guide **28** fixed to the bracket **50** is positioned in the width direction.

As shown in FIG. **6**, the stacker **1** includes a tilting mechanism **60** that lifts and tilts the stacking shelf **22** on the front F side to the lifting and lowering table **24**. That is, an end portion side (i.e., open side) of the paper S that is not provided with the stopper **26** and each paper width guide **28** is tilted upward. As shown in FIG. **7** and FIG. **8**, the tilting mechanism **60** includes a direct-moving cylinder **62** fixed to an end portion of the lifting and lowering table **24** on the front F side, and a rod **64** to be reciprocally moved in the up-down direction by the direct-moving cylinder **62**. The direct-moving cylinder **62** is electrically operated, and controlled by the stacker control unit **40** (see FIG. **4**). A tip (upper end) of the rod **64** is rotatably fixed to the stacking shelf **22** by a rotating pin **66**. A base end portion **22***a* of the stacking shelf **22** on the rear R side is rotatably fixed to the lifting and lowering table **24** by a support pin **68**. The support pin **68** is attached to an upper end of an arm part **70** disposed vertically from the lifting and lowering table **24**. By the tilting mechanism **60**, the stacking shelf **22** is rotated about the support pin **68** to be inclined to the lifting and lowering table **24**.

The lifting and lowering table **24** is a rectangular plate-shaped body in planar view, and a base end portion **24***a* in the rear R is connected to the main body **16** to be movable in the up-down direction, for example, as shown in FIG. **6**. Specifically, the lifting and lowering table **24** on a base end portion **24***a* side is fixed to a chain **72** disposed in the main body **16** via a bracket. The chain **72** has an endless state, and is hung around between sprockets **74** disposed above and below in the main body **16**. The respective sprockets **74** are arranged at opposite ends of a rotary shaft **76** extending in an axial direction in planar view of FIG. **5**. Therefore, two chains **72** are provided on each of left and right sides of the main body **16** in the width direction, and each chain **72** is provided with upper and lower sprockets **74**.

As shown in FIG. **6**, a lifting and lowering mechanism **77** is disposed in the main body **16**. The lifting and lowering mechanism **77** includes a motor **78** for the lifting and lowering table. The motor **78** for the lifting and lowering table is controlled to be forward and reverse rotatable by the stacker control unit **40**. A rotation output of the motor **78** for the lifting and lowering table is transmitted to a worm gear (lifting and lowering mechanism) **82** via a timing belt **80**. A wheel **84** is rotated by the worm gear **82**, and consequently, a spur gear **86** meshing with a teeth part of the wheel **84** rotates. The spur gear **86** is fixed to the rotary shaft **76**, and the rotary shaft **76** and the sprocket **74** are rotated by the spur gear **86**, so that the chain **72** is driven to lift and lower the lifting and lowering table **24**.

A wheel **88** is disposed on a lower side of the base end portion **24***a* of the lifting and lowering table **24**, and the wheel **88** runs along a front surface **16***a* of the main body **16**. Consequently, the lifting and lowering table **24** rises and lowers relative to the main body **16** in the cantilever state.

For example, as shown in FIG. **6**, a power receiving head (power receiving device) **90** is disposed in the front F of the stacker **1**. The power receiving head **90** is fixed to a front end **12***a* of the base **12**. One end of a power cable **92** is electrically connected to the power receiving head **90**, and the other end of the power cable **92** is electrically connected to a battery **94** (see FIG. **4**) in the main body **16**.

The battery **94** is, for example, a lithium ion battery, and includes a battery management device **97** (see FIG. **9**). The battery management device **97** manages a charged state of the battery **94**, and outputs battery information to the stacker control unit **40**.

The power receiving head **90** faces a power supply head **96** at a predetermined position such as the receiving position **PS1** (see FIG. **1**) or the paper supply position **PS2** (see FIG.

2). The power receiving head **90** receives power supplied from the power supply head **96**, for example, in a noncontact manner. The power supply head **96** is installed at a position corresponding to a stopped position of the stacker **1** to the printer **3** or the paper folding machine **5**. The power supply head **96** includes a power outlet **96***a*, and the power outlet **96***a* is connected to a power supply disposed in the vicinity. Note that a power supply method is not limited to the noncontact manner, and may be of a contact type. Also, a position where the power supply head **96** is installed is not limited to the vicinity of the printer **3** or the paper folding machine **5**, and the head may be suitably disposed at a predetermined position where the stacker **1** periodically stops.

As shown in FIG. **4**, the automatic guided vehicle **20** includes, on an upper surface, a communication unit **101** that performs transmission and reception with the conveyance management device **203** (see FIG. **10**) that is a superordinate device, and an ID reader **105**. The ID reader **105** reads the stacker ID **13** fixed to the lower surface of the base **12**. The communication unit **101** and the ID reader **105** are connected to an automatic conveyance control unit **103** that controls the automatic guided vehicle **20**, to transmit and receive a signal to and from the automatic conveyance control unit **103**.

In an upper part of the automatic guided vehicle **20**, a lifting and lowering platform **20***b* that rises and lowers in the up-down direction is disposed. As the lifting and lowering platform **20***b* rises, the stacker **1** is lifted up from the floor FL, and the automatic guided vehicle **20** runs in this state to convey the stacker **1** to the predetermined position. When the automatic guided vehicle **20** reaches a destination position, the lifting and lowering platform **20***b* is lowered to bring the leg part **14** of the stacker **1** into contact with the floor FL, thereby placing the stacker **1** at the predetermined position. For example, after placing the stacker **1** at the predetermined position, the automatic guided vehicle **20** retreats from below the stacker **1** to move to the next destination. The automatic guided vehicle **20** has a running schedule managed by the conveyance management device **203** (FIG. **10**), and the automatic guided vehicle **20** runs in accordance with conveyance instruction information received from the conveyance management device **203**.

FIG. **9** is a block diagram showing an example of a hardware configuration of the stacker **1**. As shown in FIG. **9**, the stacker **1** includes the stacker control unit **40**. The stacker control unit **40** includes, for example, a CPU (processor) **120**, a storage unit **121** that stores program or the like to be executed by the CPU **120**, and a main memory **122** that functions as a work area during the execution of each program. The storage unit **121** is, for example, a read only memory (ROM), a hard disk drive (HDD), a flash memory or the like.

A series of processing for achieving the aforementioned various types of control is stored as an example in a program form in the storage unit **121**, and the CPU **120** reads this program out to the main memory **122**, to execute information processing and arithmetic processing, thereby achieving various types of control. Note that the program may be applied in a form of being installed in advance in the storage unit **121**, a form of being provided in a stored state in a computer readable storage medium, a form of being delivered via a wired or wireless communication means, or the like. The computer readable storage medium is a magnetic disk, a magneto-optical disk, a CD-ROM, a DVD-ROM, a semiconductor memory or the like.

Furthermore, the stacker **1** includes the communication unit (transmitter) **18** to achieve communication with an after-mentioned stacker management device (see FIG. **10**), the printer **3**, and various processing machines (e.g., the paper folding machine **5** and a creasing machine **6**). The stacker control unit **40** is connected to the communication unit **18** via a bus, and the communication unit **18** transmits various types of information to a predetermined transmission destination based on the instruction from the stacker control unit **40**, and outputs information received from each device to the stacker control unit **40**.

For example, the communication unit **18** has a communication function to establish communication along various communication standards depending on the communication destination. As an example, the communication unit **18** communicates with the printer **3** and various processing machines (e.g., the paper folding machine **5**, the creasing machine **6** and the like) by use of short range communication such as Bluetooth (registered trademark), and communicates with a comparatively remotely disposed stacker management device **202** (see FIG. **10**) and a general printing management device **201** (see FIG. **10**) that is a superordinate system of the stacker management device **202** by use of wide area communication (e.g., wireless LAN or the like). Also, the communication unit **18** may communicate with the stacker management device **202** and the general printing management device **201** based on a specific communication protocol for use in a printing field.

Also, the stacker control unit **40** is connected to the aforementioned positioning motors **38** and **56**, the up-down moving motor **46**, the direct-moving cylinder **62** and the motor **78** for the lifting and lowering table via a bus, to control these respective parts. Specifically, the stacker control unit **40** receives the job information from the stacker management device **202** or the general printing management device **201**, and controls the various motors **38**, **46**, **56** and **78** and the direct-moving cylinder **62** based on the job information.

Further, the stacker control unit **40** is connected to the battery management device (microcomputer) **97** that manages the battery **94** via a bus. The stacker control unit **40** receives the battery information (e.g., a battery remaining capacity or the like) from the battery management device **97**, and transmits this battery information to the stacker management device **202** (see FIG. **10**) via the communication unit **18**.

FIG. **10** is a schematic diagram showing an example of an entire configuration of the printing system **200** including the stacker **1** and the automatic guided vehicle **20**.

As shown in FIG. **10**, the printing system **200** includes, as a management system **210**, the general printing management device **201**, the stacker management device **202**, the conveyance management device **203**, the printer management device **204**, and a processing machine management device **205**. The management devices **201** to **205** included in the management system **210** may include a configuration that allows intercommunication.

The printing system **200** includes the stacker **1** managed by the stacker management device **202**, the automatic guided vehicle **20** managed by the conveyance management device **203**, the printer **3** controlled by the printer management device **204**, and various processing machines managed by the processing machine management device **205**. FIG. **10** shows the paper folding machine **5** and the creasing machine **6** as examples of the processing machine.

The general printing management device **201** includes a configuration that allows communication with the stacker management device **202**, the conveyance management device **203**, the printer management device **204**, and the processing machine management device **205**, and generally manages the whole printing system **200** based on information from these management devices. Note that the general printing management device **201** will be described later in detail.

The stacker management device **202** is a management device including a configuration that allows communication with each of a plurality of stackers **1**, and that manages the respective stackers **1**. The stacker management device **202** includes, for example, stacker management information associated with stacker ID, operation status, current position information, operating information, and the battery information. The operation status indicates "an operating state" in a case where a job is assigned, and indicates "a standby state" in a case where the job is not assigned. As the current position information, position information of the stacker is registered. As this position information, coordinate information may be registered, or a current position may be specified by association with ID of the printer **3** or the processing machine in a case of receiving ejected paper or supplying paper. The operating information includes total operating time, time elapsed from previous operation, and the like. The battery information includes, for example, a battery charging rate and a battery remaining capacity. The stacker management device **202** communicates with each stacker **1** at a predetermined timing, to receive the operation status, the current position information, the operating information, and the battery information from each stacker **1**, and updates the stacker management information based on these pieces of information.

The stacker management device **202** determines one stacker **1** that executes the job based on the aforementioned stacker management information in a case of receiving the job ID and job information from the general printing management device **201**. For example, the stacker management device **202** includes a predetermined evaluation formula including, as parameters, the time elapsed from the previous operation, the battery remaining capacity, a distance between equipment designated by the job information (e.g., the printer **3**, the paper folding machine or the like) and the current position, and the like. Then, the stacker management device **202** substitutes, into the evaluation formula, the parameters of each stacker **1** indicating "the standby state" as the operation status from the stacker management information, to calculate an evaluation value. Then, the stacker having the highest evaluation value is selected as the stacker that executes the job. Note that in the evaluation formula, the parameter may be weighted depending on an importance degree.

The conveyance management device **203** is a management device that manages an operation of a plurality of automatic guided vehicles (automatic guided conveyance device) **20**. The conveyance management device **203** includes a configuration that allows the communication with each automatic guided vehicle **20**. Individual automatic guided vehicle IDs are assigned to the automatic guided vehicles **20**, respectively. Note that the conveyance management device **203** will be described later in detail.

Each of the general printing management device **201**, the conveyance management device **203** and each automatic guided vehicle **20** holds premise map information. This enables the automatic guided vehicle to move to a desired position in response to instructions from the general printing management device **201** and the conveyance management device **203**. Also, in this coordinate information, positions of the printer **3** and the respective processing machines (e.g., the paper folding machine **5**, the creasing machine **6** and the like) may be registered.

The printer management device **204** is a management device that manages the printer **3**. The printer management device **204** outputs, for example, the job information to the printer **3** in a case of receiving the job information from the general printing management device **201**. Also, in a case of receiving the job completion signal from the printer **3**, the printer management device **204** outputs the job completion signal to the general printing management device **201**. Alternatively, the printer management device **204** may store operating information, abnormality detection and the like of the printer **3**. These pieces of information are useful information during maintenance and inspection.

The processing machine management device **205** is a management device that manages the processing machine that performs a process on a downstream side of the printer **3**. FIG. **1** shows the paper folding machine **5** and the creasing machine **6** as examples of the processing machine, but the processing machine is not limited to these examples. Alternatively, the processing machine management device **205** may store the operating information, abnormality detection and the like of each processing machine.

Note that FIG. **10** illustrates a case where two stackers **1**, three automatic guided vehicles **20**, one printer **3**, one paper folding machine **5** and one creasing machine **6** are included in the printing system **200**, but the number of these devices is not limited to an aspect shown in the drawing. That is, there are not any special restrictions on the number as long as at least one device of one type is provided.

FIG. **11** is a block diagram showing an example of a hardware configuration of the general printing management device **201** according to the present embodiment. As shown in FIG. **11**, the general printing management device **201** includes computers, for example, a CPU **211**, a storage unit **212**, a main memory **213**, a communication unit **214**, an input unit **215**, and a display unit **216**.

The CPU **211** controls, for example, the whole printing system **200** with an operating system (OS) stored in the storage unit **212** connected via a bus, and executes various types of program stored in the storage unit **212** to execute various types of processing.

The storage unit **212** is, for example, a read only memory (ROM), a hard disk drive (HDD), a flash memory or the like, and stores, for example, OS for controlling the whole printing system **200**, such as Windows (registered trademark), an application for a printing operation, various data or files and the like. Also, the storage unit **212** stores program for achieving various types of processing, and various data required to achieve various types of processing.

The main memory **213** is constituted of a writable memory such as a cache memory or a random access memory (RAM), and is used, for example, as a work area where reading of execution program by the CPU **211**, writing of processing data by the execution program or the like is performed.

The communication unit **214** connects to network to communicate with the other devices, and functions as interface for transmitting and receiving information.

The input unit **215** is a user interface for a user to provide the general printing management device **201** with the instruction, for example, a keyboard, a mouse, a touch panel or the like.

The display unit **216** includes a display screen constituted of, for example, a liquid crystal display (LCD), an organic electro luminescence (EL) or the like, and displays results or the like of application software program executed by the CPU **211**.

Alternatively, the input unit **215** and the display unit **216** may be connected to the general printing management device **201** via network or the like, and include a configuration that allows a remote input operation and remote display.

Further, a hardware configuration of each of the stacker management device **202**, the conveyance management device **203**, the printer management device **204** and the processing machine management device **205** is a configuration substantially similar to the general printing management device **201**. That is, each of the management devices **202** to **205** also includes a CPU, a main memory, a storage unit, a communication unit, an input unit, a display unit and the like. Alternatively, the input unit and the display unit may include a configuration that allows the remote operation.

Next, description will be made as to functions of the general printing management device **201** according to the present embodiment. A series of processing for achieving various functions to be described later is stored, as an example, in a program form in the storage unit **212** shown in FIG. **11**, and the CPU **211** reads this program out to the main memory **213**, and executes information processing and arithmetic processing, to achieve various functions. Note that the program may be applied in a form of being installed in advance in the storage unit **212**, a form of being provided in a stored state in a computer readable storage medium, a form of being delivered via a wired or wireless communication means, or the like. The computer readable storage medium is a magnetic disk, a magneto-optical disk, a CD-ROM, a DVD-ROM, a semiconductor memory or the like.

FIG. **12** is a functional block diagram showing an example of a function included in the general printing management device **201**. As shown in FIG. **12**, the general printing management device **201** includes, for example, the storage unit **212**, a job management unit **222**, a processing unit **223**, and the communication unit **214**.

The storage unit **212** stores a job management list. The job management list is a list in which a manufacturing schedule of a printed product to be manufactured by the printing system **200** is registered. In the job management list, for example, for each job ID (job identification information) assigned to the printed product, job information including a registered manufacturing process procedure for manufacturing the printed product, job status and the like are registered.

The job information includes various types of information required for manufacturing the printed product, such as paper information and work information.

The paper information includes, for example, a paper size, a paper thickness, the number of sheets to be printed, the number of paper sheets to be included in the printed product, the number of copies of printed product to be prepared and the like.

The work information includes, for example, a manufacturing process procedure of the printed product, ID of a printing machine for use in a manufacturing process and set parameters.

For example, in a case of folding printed paper, the manufacturing process procedure of the printer **3** and the paper folding machine **5** in this order is registered. Also, set parameters such as a shelf height and a guide position are registered as the work information in association with printer ID of the printer **3**. Further, processing specification associated with paper folding machine ID of the paper folding machine **5**, offset information of the paper supply position for each paper folding machine ID and the like are registered.

Also, as the job information, for example, JDF described in a standard format in a printing technology field may be used.

As the job status, “completed”, “being executed”, “incomplete” or the like is registered for each manufacturing process (e.g., “printing”, “paper folding” or the like) of the printed product.

The job management unit **222** performs new addition, update, deletion and the like of the job management list stored in the storage unit **212**. For example, in a case where a new printed product manufacturing request is accepted via the input unit **215** (see FIG. **11**) or the communication unit **214**, a requested printed product is provided with the job ID, and the job information is registered in the job management list, to update the job management list.

Furthermore, in a case where the job completion signal is accepted via the communication unit **214**, the job management unit **222** updates the job status of the job management list based on the job completion signal. Consequently, a completed job, an incomplete job and a job that is being executed can be determined, and as to the job that is being executed, it is possible to determine to which process the job is completed. Consequently, job progress can be managed.

Additionally, the job management unit **222** may change a processing order of the job (job ID) of the printer **3** indicating the “incomplete” status in the job management list, depending on the job progress of the processing machine, such as the paper folding machine **5** or the creasing machine **6**.

For example, in a certain processing machine (e.g., the paper folding machine), in a case where the number of jobs to be processed is equal to or more than a predetermined number, or a case where the number of paper sheets to be processed is equal to or more than a predetermined number, a job execution order is changed to first execute the job that is not included in the processing machine (e.g., the paper folding machine) and that is included in the printer **3** indicating the “incomplete” status. Consequently, printed works in progress can be reduced, and production efficiency can be improved.

The processing unit **223** generates instruction information to be transmitted to each of the management devices **202** to **205** based on the job management list. The respective management devices **202** to **205** operate various devices under management based on the instruction information, to stably and smoothly proceed with a printing process in the printing system **200** based on the job information. Note that description will be made later as to a series of processing procedure to be executed by the processing unit **223**.

The communication unit **214** transmits various types of instruction information and the like generated by the processing unit **223** to a transmission destination designated by the processing unit **223**, and outputs the information received from various management devices **202** to **205** and the like to the processing unit **223**.

FIG. **13** is a functional block diagram showing an example of a function included in the conveyance management device **203** according to the present embodiment.

The conveyance management device **203** includes a storage unit **231**, an information acquisition unit **232**, a determination unit **233**, and a communication unit **234**.

The storage unit **231** stores, for example, conveyance management information associated with the automatic guided vehicle ID, operation status, current position information, operating information, battery information and the like.

The operation status indicates "an operating state" in a case where stacker conveyance is assigned at present, indicates "a standby state" in a case where the stacker conveyance is not assigned, and indicates "charging" in a case where the vehicle is being charged. As the current position information, position information of the automatic guided vehicle **20** is registered. The operating information includes, for example, total operating time, time elapsed from previous operation, and the like. The battery information includes, for example, a battery charging rate and a battery remaining capacity.

Note that various types of information included in the aforementioned conveyance management information are illustrated as an example, and part of the information may be registered, or another parameter may be additionally registered.

The information acquisition unit **232** communicates with each automatic guided vehicle **20** at a predetermined timing, and acquires the aforementioned battery information, current position information and operating information, to update the conveyance management information stored in the storage unit **231**.

The determination unit **233** determines one automatic guided vehicle **20** that conveys the stacker **1** based on the conveyance management information stored in the storage unit **231**, in a case of receiving, from the general printing management device **201**, conveyance instruction information including stacker ID of the stacker as a conveyance target, position information and conveyance destination information. For example, the conveyance management device **203** includes a predetermined evaluation formula including, as parameters, time elapsed from the previous operation, total operating time, battery remaining capacity, a distance between the position information of the stacker **1** of the conveyance target and the position information, and the like. Then, the parameters of the automatic guided vehicle **20** including the current operation status indicating "the standby state" are acquired from the conveyance management information, and substituted into the evaluation formula, to calculate an evaluation value. Then, the automatic guided vehicle **20** having the highest evaluation value is selected as the automatic guided vehicle that executes the conveyance instruction information. Note that in the evaluation formula, the parameter may be weighted depending on an importance degree.

The communication unit **234** establishes communication with the automatic guided vehicle **20** and communication with the general printing management device **201**, to achieve intercommunication.

Next, description will be made as to printed product manufacturing management processing to be executed by the management system **210** including the general printing management device **201**, with reference to the drawings.

FIG. **14** is a flowchart mainly showing an example of a procedure of processing to be executed by the general printing management device **201**, the stacker management device **202**, and the conveyance management device **203** in the printed product manufacturing management processing of the management system **210** concerning the printing process. FIG. **15** is a flowchart mainly showing an example of a procedure of processing to be executed by the general printing management device **201**, the stacker management device **202**, and the conveyance management device **203** in the printed product manufacturing management processing of the management system **210** concerning a processing process.

As shown in FIG. **14**, first, the general printing management device **201** determines the job ID to start manufacturing of a printed product based on the job management list (SA**1**). Then, the device transmits the determined job ID and job information associated with the job ID to the stacker management device **202** and the printer management device **204** (see FIG. **10**) (SA**2**). The printer management device **204** transmits the received job ID and job information to the printer **3**. The printer **3** receiving the job ID and job information is in the standby state until receiving a preparation completion signal from the stacker **1**.

On the other hand, on receiving the job ID and job information from the general printing management device **201**, the stacker management device **202** determines the stacker **1** that executes the job, based on the stacker management information (SA**3**), and associates the stacker information including the stacker ID and current position information of the determined stacker **1** with the job ID to transmit the associated information to the general printing management device **201** (SA**4**). Also, the stacker management device **202** transmits the job ID and job information to the determined stacker **1**. Further, the stacker management device **202** changes the operation status of the stacker **1** to which the job is assigned to "the operating state" in the stacker management information.

The stacker control unit **40** (see FIG. **9**) of the stacker **1** receiving the job ID and job information controls the motor **78** for the lifting and lowering table based on the job information. Consequently, the motor **78** for the lifting and lowering table operates, and positions the lifting and lowering table at a height position individually set depending on the type of printer **3**. Therefore, it is possible to appropriately receive the paper ejected from the printer **3** during the printing. Also, the stacker control unit **40** controls the positioning motors **38** and **56** and the up-down moving motor **46**. Consequently, a position of the stopper **26** in a front-rear direction and a position of the paper width guide **28** in the width direction are determined at positions corresponding to a size of the paper S that is described in a printing job.

On the other hand, in FIG. **14**, on receiving the stacker information and job ID from the stacker management device **202**, the general printing management device **201** generates the conveyance instruction information based on the received stacker information, job ID and job information, to transmit the information to the conveyance management device **203** (SA**5**). The conveyance instruction information includes the current position information of the stacker **1**, the stacker ID, and information of the receiving position PS**1** of the printer **3**.

The conveyance management device **203** determines the automatic guided vehicle **20** that conveys the stacker **1**, based on the conveyance instruction information and conveyance management information, and transmits the conveyance instruction information to the determined automatic guided vehicle **20** (SA**6**). Also, the conveyance management device **203** changes the operation status of the automatic guided vehicle **20** to which the conveyance instruction is assigned to "the operating state" in the conveyance management information.

The automatic guided vehicle **20** receiving the conveyance instruction information moves the stacker **1** to the receiving position PS**1** of the printer **3** based on the conveyance instruction information. Note that when the automatic guided vehicle **20** reaches the position of the stacker **1**, the stacker ID **13** may be read by the ID reader **105**, and it may be checked whether or not the stacker ID included in the conveyance instruction information matches the stacker ID read by the ID reader **105**. Thus, the checking is performed, so that the stacker **1** that is the target of the conveyance instruction can be securely moved, for example, even in a case where a plurality of stackers **1** are arranged close to each other.

On placing the stacker **1** at the receiving position PS**1** of the printer **3** based on the conveyance instruction information, the automatic guided vehicle **20** transmits a conveyance completion signal to the stacker **1**, and transmits the conveyance completion signal and the automatic guided vehicle ID of its own to the conveyance management device **203**. Note that the automatic guided vehicle **20** may transmit the conveyance completion signal to the stacker **1** via the conveyance management device **203** and the stacker management device **202**. Alternatively, communication between the automatic guided vehicle **20** and the stacker **1** to be described hereinafter may be directly mutually performed, or may be indirectly mutually performed via the conveyance management device **203** and the stacker management device **202**.

On receiving the conveyance completion signal and the automatic guided vehicle ID (SA**7**), the conveyance management device **203** transmits the conveyance completion signal to the general printing management device **201** (SA**8**). Furthermore, the conveyance management device **203** acquires the battery information of the automatic guided vehicle **20** receiving the conveyance completion signal, and determines whether or not the battery remaining capacity is equal to or less than a predetermined lower limit value. As a result, in a case where the battery remaining capacity is equal to or less than the lower limit value, the device transmits, to the automatic guided vehicle **20**, charging instruction information for guiding the vehicle to a battery station, and changes the operation status of the conveyance management information to "a charging state". Further, in a case where the battery remaining capacity is in excess of the lower limit value, the operation status is changed to "a standby state".

On the other hand, for example, the stacker **1** receiving the conveyance completion signal from the automatic guided vehicle **20** transmits the preparation completion signal to the printer **3**. Also, the battery **94** (see FIG. **9**) of the stacker **1** receives power supply from the power supply head **96** disposed in the vicinity of the printer **3** via the power receiving head **90** as required.

On receiving the preparation completion signal from the stacker **1**, the printer **3** starts printing based on the job information received from the printer management device **204**. The printer **3** transmits a printing start signal to the printer management device **204**. The printer management device **204** manages a status. Alternatively, the printer management device **204** may transmit the printing start signal to the general printing management device **201**.

Near the paper ejection port **3***b* of the printer **3**, a sensor that detects ejected paper is disposed. The printer **3** counts the number of sheets to be printed based on a detection signal from the sensor, and transmits the count number to the stacker **1**. The stacker control unit **40** of the stacker **1** controls the motor **78** for the lifting and lowering table based on the count number and the paper thickness acquired from the job information. Consequently, the lifting and lowering table can lower depending on the number of stacked sheets, and receive the paper ejected from the printer **3** at an appropriate position.

On receiving that the count number reaches the number of sheets to be printed prescribed in the job information, the printer **3** transmits a printing job completion signal to the stacker **1** disposed at the receiving position PS**1** and the printer management device **204**.

On receiving the printing job completion signal, the stacker control unit **40** (see FIG. **9**) of the stacker **1** controls the motor **78** for the lifting and lowering table, and lowers the lifting and lowering table to a position during moving. Furthermore, the stacker control unit **40** controls the direct-moving cylinder **62**, to place the stacking shelf **22** in an inclined state at a predetermined angle, and prepares for the next process of conveyance to the processing machine. For example, as shown in FIG. **7** and FIG. **8**, the tilting mechanism **60** of the stacker **1** brings a state where the front F side of the stacking shelf **22** is disposed above the rear R side. Consequently, when the automatic guided vehicle **20** conveys the stacker **1** toward the front F, the paper S stacked on the stacking shelf **22** can be prevented from scattering and falling from the open side that is not provided the stopper **26** and the respective paper width guides **28**.

On the other hand, on receiving the printing job completion signal from the printer **3**, the printer management device **204** transmits the printer ID, job ID and printing job completion signal to the general printing management device **201**.

On receiving the printing job completion signal or the like (SA**9**), the general printing management device **201** changes a printing process status of the job ID of the job management list to "completed" to update the job management list (SA**10**). Then, the device returns to the step SA**1**, and determines the job ID to be next executed from the job management list. Consequently, a printing process of the job ID to be next executed is executed, and the aforementioned processing of the determined job is performed.

Furthermore, on receiving the printing job completion signal as described above (SB**1** of FIG. **15**), the general printing management device **201** specifies the processing machine (e.g., the paper folding machine **5**) that executes the processing process from the job information associated with the job ID that receives the printing job completion signal, and transmits processing machine ID, job ID and job information to the processing machine management device **205** (SB**2**). Note that at this time, stacker ID of the stacker **1** that supplies paper to the processing machine may be transmitted together.

The processing machine management device **205** that receives these pieces of information transmits the job ID, job information and stacker ID to the processing machine (e.g., the paper folding machine **5**) specified from the processing machine ID. For example, on receiving the job ID and job information, the paper folding machine **5** changes setting based on the job information, and is in the standby state for the job until the stacker **1** moves to be disposed at the paper supply position PS**2**.

Furthermore, the general printing management device **201** generates conveyance instruction information for moving the stacker **1** disposed at the receiving position PS**1** of the printer **3** to the paper supply position of the paper folding machine **5**, and transmits the information to the conveyance management device **203** (SB**3**). The conveyance instruction information may include stacked paper information (e.g., a paper size, paper thickness, and the number of paper sheets) of the stacker ID in addition to the stacker ID and current position information of the stacker. Also, the conveyance instruction information may include offset information to the paper supply position of the paper folding machine **5**. Alternatively, the conveyance instruction information may include information concerning a paper supply direction to the paper folding machine **5** (e.g., a vertical direction or a lateral direction or the like). Alternatively, as the current position information of the stacker, position information of the receiving position PS**1** of the printer **3** may be used.

The conveyance management device **203** determines the automatic guided vehicle **20** that conveys the stacker **1**, based on the conveyance instruction information and conveyance management information, and transmits the conveyance instruction information to the determined automatic guided vehicle **20** (SB**4**). The automatic guided vehicle **20** receiving the conveyance instruction information moves the stacker **1** from the receiving position PS**1** of the printer **3** to the paper supply position PS**2** of the paper folding machine **5** based on the conveyance instruction information. Note that when the automatic guided vehicle **20** reaches the position of the stacker **1**, the stacker ID **13** may be read by the ID reader, and it may be checked whether or not the stacker ID included in the conveyance instruction information matches the stacker ID read by the ID reader **105**.

Furthermore, during the conveyance by the automatic guided vehicle **20**, the stacking shelf **22** in the stacker **1** is inclined at the predetermined angle. Consequently, the paper S stacked on the stacking shelf **22** can be prevented from scattering and falling from the open side that is not provided with the stopper **26** and the respective paper width guides **28** during running, and it is possible to achieve stable running.

Alternatively, in the automatic guided vehicle **20**, acceleration or speed during the conveyance may be adjusted depending on the paper information included in the conveyance instruction information. For example, a stacking weight of paper to be stacked by the stacker **1** can be estimated based on the number of the paper sheets, paper size and paper thickness. In the automatic guided vehicle **20**, the acceleration or speed is adjusted depending on a total weight of the stacker **1** that is obtained from these pieces of information, to achieve the stable running. Note that the conveyance management device **203** may estimate the stacking weight of the paper from the paper information such as the number of the paper sheets, and the conveyance instruction information to be transmitted to the automatic guided vehicle **20** may include estimated stacking information.

The automatic guided vehicle **20** moves the stacker **1** to the paper supply position PS**2** of the paper folding machine **5**, and controls orientation of the stacker **1** so that the paper supply direction of the stacker **1** is appropriate, based on information concerning the paper supply direction to the paper folding machine **5**, the information being included in the conveyance instruction information. Consequently, even in a case where an ejection orientation of the sheet in the printer **3** is different from a paper supply orientation of the sheet in the paper folding machine **5**, the stacker **1** can be installed with an appropriate orientation to obtain an appropriate paper supply orientation depending on the paper supply orientation in the paper folding machine **5**.

Furthermore, the automatic guided vehicle **20** adjusts a position of a center line CL**1** of the stacker **1** to a center line CL**2** of the paper folding machine **5** in planar view, based on the offset information included in the conveyance instruction information. For example, a positional relation between the center line CL**1** of the stacker **1** and the center line CL**2** of the paper folding machine **5** varies depending on whether a desired folding position is a center of the paper or offset from the center of the paper, relative to the paper folding machine **5** in which a fixing position of a knife (not shown) for folding the paper is on the center line CL**2**.

For example, in a case where the desired folding position is offset from the center of the paper relative to the paper folding machine **5** in which the position of the knife is on the center line CL**2**, it is necessary to align the center line CL**1** of the stacker **1** at the paper supply position PS**2** depending on offset. Thus, the paper supply position PS**2** varies with the paper folding machine **5**, paper size, desired folding position or the like, and hence the conveyance instruction information to be transmitted to the automatic guided vehicle **20** includes the offset information as information concerning the paper supply position PS**2**.

For example, the automatic guided vehicle **20** places the stacker **1** at a position at which the center line CL**2** of the paper folding machine **5** coincides with the center line CL**1** of the stacker **1** in planar view as shown in FIG. **16**, in a case where offset is zero. Also, the automatic guided vehicle **20** places the stacker **1** at a position at which the center line CL**1** of the stacker **1** is offset from the center line CL**2** of the paper folding machine **5** as shown in FIG. **17**, depending on the offset in a case where the offset is not zero.

On placing the stacker **1** at the paper supply position PS**2** depending on the offset information, the automatic guided vehicle **20** transmits the conveyance completion signal to the stacker **1**, and transmits the conveyance completion signal and the automatic guided vehicle ID of its own to the conveyance management device **203**.

On receiving the conveyance completion signal and automatic guided vehicle ID (SB**5**), the conveyance management device **203** transmits the conveyance completion signal to the general printing management device **201** (SB**6**).

Further, the conveyance management device **203** acquires the battery information of the automatic guided vehicle **20** receiving the conveyance completion signal, and determines whether or not a battery remaining capacity is equal to or less than a predetermined lower limit value. As a result, in a case where the battery remaining capacity is equal to or less than the lower limit value, the device transmits, to the automatic guided vehicle **20**, charging instruction information for guiding the vehicle to a battery station, and changes the operation status of the conveyance management information to "a charging state". Further, in a case where the battery remaining capacity is in excess of the lower limit value, the operation status is changed to "a standby state".

On the other hand, the stacker control unit **40** of the stacker **1** receiving the conveyance completion signal from the automatic guided vehicle **20** controls the direct-moving cylinder **62**, to return the stacking shelf **22** to a horizontal state. Furthermore, the stacker control unit **40** acquires information in the processing process (e.g., height information of the lifting and lowering table or the like) from the job information already received from the stacker management device **202**, and controls the up-down moving motor **46** and the motor **78** for the lifting and lowering table based on the acquired information.

Consequently, the motor **78** for the lifting and lowering table is operated, and the stacking shelf **22** is positioned at a height position of supplied paper that is individually set depending on the type of paper folding machine **5**. Furthermore, the stacker control unit **40** operates the up-down moving motor **46** (see FIG. **6**), and displaces the stopper **26** of the stacker **1** downward as shown in FIG. **18**. This avoids interference of a paper separator **5***a* of the paper folding machine **5** with the stopper **26**.

Also, the battery **94** of the stacker **1** receives power supply from the power supply head **96** disposed in the vicinity of the paper folding machine **5** via the power receiving head **90** as required.

On completing the positioning and obtaining a paper suppliable state to the paper folding machine **5**, the stacker control unit **40** transmits a preparation completion signal to the paper folding machine **5**. On receiving the preparation completion signal from the stacker **1**, the paper folding machine **5** determines whether or not an upper surface of the lifting and lowering table of the stacker **1** is detected by an upper surface detection sensor (not shown) disposed near a paper supply port of the paper folding machine **5**.

As a result, in a case where the upper surface is not detected, the paper folding machine **5** transmits an instruction to the stacker **1** to raise the lifting and lowering table. Consequently, the stacker control unit **40** controls the motor **78** for the lifting and lowering table, and raises the lifting and lowering table. This operation is performed until the upper surface of the lifting and lowering table is detected by the upper surface detection sensor. Then, when the upper surface of the lifting and lowering table is detected by the upper surface detection sensor, the paper folding machine **5** determines that the lifting and lowering table of the stacker **1** is disposed at an appropriate position, and starts folding paper based on the job information received from the processing machine management device **205**.

Alternatively, the paper folding machine **5** may acquire the stacker ID of the stacker **1**, and perform check processing of checking whether or not the acquired stacker ID matches the stacker ID associated with the job ID to be started from now on, before starting the job. Thus, the checking is performed, so that it can be confirmed whether or not the stacker matching the job to be started from now on is disposed at the paper supply position PS**2**.

Near a paper ejection port of the paper folding machine **5**, a sensor that detects ejected paper is disposed. The paper folding machine **5** counts the number of sheets to be processed based on a detection signal from the sensor. The number of the sheets to be processed is transmitted to the stacker **1** directly from the paper folding machine **5** or via the processing machine management device **205** and the stacker management device **202**. The stacker control unit **40** controls the motor **78** for the lifting and lowering table depending on a relation between the number of remaining sheets and a height of a paper supply unit of the paper folding machine **5**, and moves the lifting and lowering table to an appropriate height position in accordance with proceeding of processing.

Then, on detecting that the count number reaches the number of the sheets to be processed that is prescribed in the job information, the paper folding machine **5** transmits a processing job completion signal to the stacker **1** disposed at the paper supply position PS**2** and the processing machine management device **205**.

On receiving the processing job completion signal, the stacker control unit **40** of the stacker **1** controls the motor **78** for the lifting and lowering table, and lowers the lifting and lowering table to the position during the moving. Also, the stacker **1** transmits the processing job completion signal and the stacker ID of its own to the stacker management device **202**. On receiving the processing job completion signal (SB**7**), the stacker management device **202** changes the operation status of the received stacker ID to "the standby state" (SB**8**).

On the other hand, on receiving the processing job completion signal from the paper folding machine **5**, the processing machine management device **205** transmits the paper folding machine ID, job ID and processing job completion signal to the general printing management device **201**.

On receiving the processing job completion signal and the like (SB**9**), the general printing management device **201** changes, to "completed", a status of the processing process of the job ID of the job management list, to update the job management list (SB**10**). Then, the device returns to the step SB**1**, and is in the standby state until receiving the next printing job completion signal. Also, in a case where the printing job completion signal is already received, the subsequent processing is executed.

As described above, according to the present embodiment, the following operations and effects are exhibited.

For example, the general printing management device **201** generates the conveyance instruction information for conveying the stacker **1** in which the paper S ejected from the printer **3** is stacked from the receiving position PS**1** of the printer **3** to the paper supply position PS**2** of the next processing machine (e.g., the paper folding machine **5**), and transmits the information to the conveyance management device **203**. Consequently, by use of the automatic guided vehicle **20**, it is possible to automatically move the stacker **1** from the printer **3** to the processing machine in a next process, and it is possible to achieve personnel saving.

Furthermore, the conveyance instruction information includes at least one of the sheet size, the sheet thickness, or the number of the sheets to be stacked in the stacker **1**, and hence the weight of the stacker **1** can be estimated based on the information. Consequently, it is possible to select the appropriate automatic guided vehicle **20** for conveying the stacker **1** in consideration of the weight of the stacker. Furthermore, during the conveying of the stacker **1**, an appropriate speed or acceleration is set based on paper information, so that it is possible to achieve the stable running.

Additionally, the conveyance instruction information includes processing machine identification information individually given to the processing machine and the offset information of the sheet supply position in the processing machine, and hence it is possible to install the stacker **1** at an appropriate sheet supply position depending on the desired folding position even in a case where the paper supply position of the sheet varies with a type of processing machine, paper size, and quire shape. This can achieve stable and smooth sheet supply.

In addition, the conveyance instruction information for conveying the stacker **1** in which the sheet is not stacked to the receiving position PS**1** of the printer **3** is generated based on the job information, so that it is possible to automatically move the stacker **1** to the printer **3**, and it is possible to achieve further personnel saving.

Furthermore, the conveyance instruction information includes the stacker ID, and hence, for example, even in a case where a plurality of stackers **1** are installed close to one another, the stackers are checked based on the stacker ID, and it is therefore possible to prevent incorrect conveyance indicating that the stacker **1** that is not a conveyance target is incorrectly conveyed.

Additionally, since the conveyance instruction information includes the information concerning the supply direction of the paper S to the processing machine, it is possible to supply the paper S in an appropriate orientation depending on processing specifications in each processing machine, even in a case where an ejection orientation of the paper S in the printer **3** is different from a supply orientation of the paper S in the processing machine.

Description has been made above as to the present disclosure by use of the embodiment, but technical scope of the present disclosure is not restricted by the description of the embodiment. The embodiment can be variously changed or modified without departing from the scope of the disclosure, and even a changed or modified aspect is included in the technical scope of the present disclosure.

Furthermore, the flow of various processes described above in the embodiment is merely an example, and unnecessary steps may be deleted, new steps may be added and the processing order may be changed without departing from the scope of the present disclosure.

In the aforementioned embodiment, communication is performed between the general printing management device **201** and each of the management devices **202** to **205**, but the functions of the management devices **202** to **205** may be included in the general printing management device **201**. Also, the communication between the automatic guided vehicle **20** and the general printing management device **201** is performed via the conveyance management device **203**, but the automatic guided vehicle **20** and the general printing management device **201** may directly transmit and receive information without passing through the conveyance management device **203**. This also applies to the other management device. For example, the stacker **1**, the printer **3**, various processing machines and the general printing management device **201** may directly transmit and receive information.

Furthermore, one of the management devices may include the function of the other management device. For example, the processing machine management device **205** may include the functions of the stacker management device **202** and the conveyance management device **203**.

Modification 1

In the aforementioned embodiment, the paper folding machine **5** is illustrated and described as the processing machine to which paper is supplied from the stacker **1**, but the present disclosure is not limited to this example. For example, as shown in FIG. **19**, the present disclosure can be applied also to the creasing machine **6** as the processing machine. That is, in a case where the creasing machine **6** is registered as a post-process of the printer **3** in the job information, processing similar to the aforementioned processing is performed, so that paper can be stably supplied also to the creasing machine **6**.

Also, in the case of the creasing machine **6**, for example, the stacker control unit **40** of the stacker **1** may control the motor **78** for the lifting and lowering table depending on a relation between the number of sheets of paper S to be stacked and a height of a paper supply unit of the creasing machine **6**, to adjust the lifting and lowering table to an appropriate height position depending on proceeding of processing.

Modification 2

In the aforementioned embodiment, the stacker **1** is moved by the automatic guided vehicle **20**, but the present disclosure is not limited to this example. For example, as shown in FIG. **20** and FIG. **21**, each of leg parts **14** of some of stackers **1** may include a caster **110** so that the stacker **1** can run without using the automatic guided vehicle **20**. In this case, a handle **112** is disposed on an upper part of a back surface of the main body **16**, and a worker conveys the stacker **1**. The stacker **1** is positioned by a stopper **110*a*** for the caster (see FIG. **21**) attached to the caster **110** at a fixed position such as the receiving position PS**1** or the paper supply position PS**2**.

In this case, the stacker ID **13** is disposed on a lower surface of the base **12** on the front F side, and an ID reader **114** is installed in the vicinity of the printer **3** or the processing machine such as the paper folding machine **5**. Data received by the ID reader **114** is transmitted to a control unit of the printer **3** or the processing machine, such as the paper folding machine **5**.

Note that the stacker **1** may include a running device such as a running motor, and run by itself.

Also, a tilt angle of the stacking shelf **22** during the running of the stacker **1** may be changed depending on a stacking amount of the paper S. For example, the tilt angle when the stacking amount is small may be larger than the tilt angle when the stacking amount is large.

The tilt angle of the stacking shelf **22** may be changed depending on a magnitude of acceleration (including deceleration that is negative acceleration) during the running of the stacker **1**. For example, the tilt angle when the acceleration of the stacker **1** is large may be larger than the tilt angle when the acceleration is small.

Modification 3

The conveyance instruction information transmitted in step SB**3** or SB**4** (see FIG. **15**) may include the orientation of the paper S to be ejected from the printer **3** to the stacker **1**, that is, information concerning a stacking direction of the paper S in the stacker **1**. For example, the conveyance instruction information may include information indicating whether the paper S is longitudinally stacked or laterally stacked in the stacker **1**.

The automatic guided vehicle **20** may convey the stacker **1** so that a longitudinal direction of the paper S is a traveling direction, based on information concerning the orientation of the paper S that is included in the conveyance instruction information.

For example, the automatic guided vehicle **20** may include a rotating mechanism that rotates the stacker **1** about a vertical axis, and the stacking direction of the stacker **1** to the traveling direction of the automatic guided vehicle **20** may be rotated about the vertical axis by the rotating mechanism. The rotating mechanism may include a rotating table disposed in an upper part of the automatic guided vehicle **20** to support the stacker **1**, a rotary shaft that rotatably supports the rotating table about the vertical axis to a main body of the automatic guided vehicle **20**, a rotating motor that rotates the rotary shaft about the vertical axis, and the like.

Consequently, since the conveyance instruction information includes the information concerning the stacking direction of the paper S in the stacker **1**, the automatic guided vehicle **20** can convey the stacker **1** so that the longitudinal direction of the paper S is the same as the traveling direction. Consequently, it is possible to prevent sheet collapse during the conveying.

Modification 4

In the conveyance management device **203**, the conveyance instruction information to be transmitted to the automatic guided vehicle **20** may include information of a running route along which the automatic guided vehicle **20** runs and information of a particularity on the running route. The particularity includes at least one of floor slope information, floor step information, temperature, humidity, air conditioning air volume, or air conditioning wind direction.

In the automatic guided vehicle **20**, a speed or acceleration may be adjusted, or the running route may be reconstructed based on the information of the particularity on the running route that is included in the conveyance instruction information.

For example, in a case where there is a slope at an angle equal to or more than a predetermined angle in a direction crossing a traveling direction, in the automatic guided vehicle **20**, a route that detours a slope section may be reconstructed, or an orientation of the stacker **1** may be turned relative to the traveling direction so that the longitudinal direction of the paper S stacked in the stacker **1** is the same as a slope direction before the slope section.

Furthermore, in a case where there is a descending slope in the traveling direction, in the automatic guided vehicle **20**, a detour route may be reconstructed to avoid the descending slope. Additionally, as shown in FIG. **8**, in a case of running in a state where the stacking shelf **22** in the stacker **1** is tilted at a predetermined angle, the automatic guided vehicle **20** may run on a descending slope section, after stacking tilt of the paper S is adjusted to decrease a difference between a descending slope angle and a tilt angle of the paper S before the slope section. For example, the automatic guided vehicle **20** may run on the descending slope section, after the difference between the descending slope angle and the tilt angle of the paper S is decreased by rotating the orientation of the stacker **1** by 180 degrees.

Additionally, in a case where there is the slope on the running route, the vehicle may run at a speed lower than a usual speed in the slope section.

In addition, in a case where there is a step on a floor, for example, in a state where the paper S is stacked in the stacker **1**, the detour route may be reconstructed, and in a state where the paper S is not stacked, the vehicle may pass as it is.

Furthermore, the paper S is easily influenced by temperature and humidity, for example, moisture absorption in high humidity, condensation due to sudden change in temperature, or static electricity or paper powder scattering in low humidity. In general, the temperature and humidity are adjusted in a printing site. However, for example, in a case where the running route included in the conveyance instruction information received from the conveyance management device **203** includes movement across buildings (movement from a paper warehouse to a processing site), the automatic guided vehicle **20** might pass through an area where the temperature and humidity are not adjusted. In this case, for example, in the state where the paper S is stacked in the stacker **1**, the running route may be reconstructed so that the vehicle moves only in an area where the temperature and humidity are adjusted. Furthermore, in the state where the paper S is not stacked in the stacker **1**, the vehicle may pass through the area where the temperature and humidity are not adjusted without changing the route. Furthermore, for example, there is also the paper S that is hard to be influenced by the temperature and humidity (e.g., a resin film or the like), and hence it may be determined whether or not to change the route, depending on a type or state of paper S stacked in the stacker **1**.

Additionally, in a case where the air conditioning air volume is larger than a predetermined value, the route may be reconstructed to detour a section where the air volume is large, or the stacker **1** may be rotated to orient the main body **16** of the stacker **1** in a windward direction before the section.

Thus, the conveyance instruction information includes the running route along which the stacker **1** is moved from a conveyance source to a conveyance destination and the information of the particularity on the running route, and hence in the automatic guided vehicle **20**, adjustment of a conveyance speed or acceleration, reconstruction of the running route and adjustment of the orientation of the stacker **1** can be performed depending on floor slope or the like. Consequently, it is possible to inhibit load collapse of the paper S during the moving.

Modification 5

In the above described embodiment, it has been illustrated the case where one stacker **1** is required for one job, but a plurality of stackers **1** may be required to execute one job. For example, in a case where the number of paper sheets in one job is set to 5000 and a maximum stacking capacity of the stacker **1** is 3000 sheets, two stackers **1** are required. In this case, the stacker management device **202** determines a plurality of stackers **1** that execute the job. The general printing management device **201** generates conveyance instruction information based on information of the plurality of stackers **1** determined by the stacker management device **202**. At this time, the conveyance instruction information may include stacker ID of the plurality of stackers **1** that execute one job.

Furthermore, depending on the job, paper may be supplied to the next processing machine in order from paper last ejected from the printer **3**. In this case, it is necessary to change (reverse) an order of the stackers **1** in which the paper S ejected from the printer **3** is stacked, and an order of the stackers **1** that supply paper to the next processing machine. To deal with this case, the conveyance instruction information transmitted from the general printing management device **201** may include at least one of the order of the stackers **1** in which the paper S ejected from the printer **3** is stacked or the order of the stackers **1** that supply the paper S to the next processing machine. Consequently, even in a case where the order of the stackers **1** in which the paper S ejected from the printer **3** is stacked is different from the order of the stackers **1** that supply paper to the next processing machine, it is possible to smoothly supply the paper S.

Such information is included in the conveyance instruction information, and hence in a case where the number of the paper sheets for one job is in excess of a maximum number of sheets to be stacked in the stacker **1**, it is possible to smoothly execute the job.

In the aforementioned embodiment and the modifications, paper has been described as an example of a medium to be conveyed by the stacker **1**, but the present invention is not limited to this example. For example, the present invention may be applied to a sheet-shaped medium such as a resin film.

Also, a rotating mechanism that rotates the stacking shelf **22** about a vertical axis may be disposed, and the stacking shelf **22** may be rotated about the vertical axis by the stacker control unit **40**. For example, the rotating mechanism is disposed between the stacking shelf **22** and the lifting and lowering table **24**, and includes a rotary shaft that rotatably supports the stacking shelf **22** around the vertical axis to the lifting and lowering table **24**, a rotating motor that rotates the stacking shelf **22** about the rotary shaft, and the like.

Since the rotating mechanism is included, the paper S is received from the printer **3**, and the stacking shelf **22** is then rotated by 90° to access a sheet processing machine such as the paper folding machine **5**. The paper S can be supplied in a rotated state by 90° from a received state. Furthermore, as described above, in place of the rotating mechanism that rotates the stacking shelf **22**, the automatic guided vehicle **20** may rotate the whole stacker **1** to change the orientation of the paper S.

REFERENCE SIGNS LIST

1 stacker (sheet stacking apparatus)
3 printer
3*a* back surface
3*b* paper ejection port
5 paper folding machine (processing machine)
5*a* paper separator
6 creasing machine (processing machine)
7 communication unit
10 shelf unit
12 base
12*a* front end
13 stacker ID (identification information)
14 leg part
16 main body
16*a* front surface
18 communication unit
20 automatic guided vehicle (automatic guided conveying apparatus)
20*a* wheel
20*b* lifting and lowering platform
22 stacking shelf
22*a* base end portion
24 lifting and lowering table
24*a* base end portion
26 stopper
26*a* upper and lower rack
28 paper width guide
30 stopper running groove
32 bracket
34 slide guide shaft
36 feed screw
38 positioning motor
40 stacker control unit
42 pinion gear
44 rotary shaft
46 up-down moving motor
48 paper width guide running groove
50 bracket
52 slide guide shaft
54 feed screw
56 positioning motor
60 tilting mechanism
62 direct-moving cylinder
64 rod
66 rotating pin
68 support pin
70 arm part
72 chain
74 sprocket
76 rotary shaft
77 lifting and lowering mechanism
78 motor for lifting and lowering table
80 timing belt
82 worm gear (lifting and lowering mechanism)
84 wheel
86 spur gear
88 wheel
90 power receiving head (power receiving device)
92 power cable
94 battery
96 power supply head
96*a* power outlet
97 battery management device
101 communication unit (transmitter)
103 automatic conveyance control unit
105 ID reader
110 caster
110*a* stopper for caster
112 handle
114 ID reader
120 CPU (processor)
121 storage unit (auxiliary memory)
122 main memory
200 printing system
201 general printing management device
202 stacker management device
203 conveyance management device
204 printer management device
205 processing machine management device
210 management system
211 CPU (processor)
212 storage unit (auxiliary memory)
213 main memory
214 communication unit (transmitter)
215 input unit
216 display unit
222 job management unit
223 processing unit
231 storage unit (auxiliary memory)
232 information acquisition unit
233 determination unit
234 communication unit (transmitter)
CL**1** center line (of stacker)
CL**2** center line (of folding machine)
F front (of base)
FL floor
PS**1** receiving position
PS**2** paper supply position (supply position)
R rear (of base)
S paper

What is claimed is:

1. A general printing management device connected to be communicable with a conveyance management device that manages a plurality of automatic guided conveying apparatuses each conveying a sheet stacking apparatus in which a sheet ejected from a printer is stackable, the general printing management device comprising:

a processor configured to generate a conveyance instruction information for conveying, by one of the automatic guided conveying apparatuses, the sheet stacking apparatus in which the sheet ejected from the printer is stacked from a sheet receiving position of the printer to a sheet supply position of a next processing machine, based on a job information in which a manufacturing process procedure for manufacturing a printed product is registered; and

a transmitter configured to transmit the conveyance instruction information to the conveyance management device,

wherein the conveyance instruction information includes information concerning an orientation of a sheet to be ejected from the printer to the sheet stacking apparatus.

2. The general printing management device according to claim 1, wherein the conveyance instruction information includes at least one of a sheet size, a sheet thickness, or a number of sheets to be stacked in the sheet stacking apparatus.

3. The general printing management device according to claim 1, wherein the conveyance instruction information includes processing machine identification information individually given to the processing machine and an offset information of the sheet supply position in the processing machine.

4. The general printing management device according to claim 1, wherein the conveyance instruction information includes information concerning a paper supply orientation of a stacker to the processing machine.

5. The general printing management device according to claim 1, wherein, in response to a plurality of sheet stacking apparatuses being required for execution of one job, the conveyance instruction information includes an identification information of the plurality of sheet stacking apparatuses that execute the one job, and at least one of an order of the sheet stacking apparatuses in which sheets ejected from the printer are stacked or an order of the sheet stacking apparatuses that supply the sheets to the next processing machine.

6. The general printing management device according to claim 1, wherein the processor is configured to generate, based on the job information, conveyance instruction information for conveying the sheet stacking apparatus in which the sheet is not stacked to the sheet receiving position of the printer.

7. The general printing management device according to claim 1, wherein the conveyance instruction information includes an identification information individually given to the sheet stacking apparatus.

8. A printing system comprising:

the general printing management device according to claim 1, and

the conveyance management device, which is configured to manage a plurality of automatic guided conveying apparatuses each conveying a sheet stacking apparatus in which a sheet ejected from a printer is stackable.

9. A conveyance management device configured to manage a plurality of automatic guided conveying apparatuses each conveying a sheet stacking apparatus in which a sheet ejected from a printer is stackable, the conveyance management device comprising:

a processor; and

a memory including a program that, when executed by the processor, causes the processor to perform operations, the operations including:

acquiring at least one of a battery information, an operating information, or a current position information of each of the plurality of automatic guided conveying apparatuses;

receiving a conveyance instruction information for conveying the sheet stacking apparatus;

determining one automatic guided conveying apparatus based on the information acquired by the acquired information and the conveyance instruction information; and

transmitting the conveyance instruction information to a determined automatic guided conveying apparatus, the conveyance instruction information including at least one of a sheet size, a sheet thickness and a number of sheets to be stacked in the sheet stacking apparatus, an identification information of the sheet stacking apparatus, or an identification information of the printer or a processing machine that is a conveyance destination of the sheet stacking apparatus,

wherein the conveyance instruction information includes information concerning an orientation of a sheet to be ejected from the printer to the sheet stacking apparatus.

10. The conveyance management device according to claim 9, wherein the conveyance instruction information includes an information concerning a running route from a conveyance source to a conveyance destination of the sheet stacking apparatus and an information of a particularity on the running route, and

wherein the particularity includes at least one of a floor slope information, a floor step information, a temperature, a humidity, an air conditioning air volume, or an air conditioning wind direction.

11. A method, comprising:

generating a conveyance instruction information for conveying, by one of a plurality of automatic guided conveying apparatuses, a sheet stacking apparatus in which a sheet ejected from a printer is stacked to a sheet supply position of a next processing machine, based on a job information in which a manufacturing process procedure for manufacturing a printed product is registered, wherein the conveyance instruction information includes information concerning an orientation of a sheet to be ejected from the printer to the sheet stacking apparatus; and

transmitting the conveyance instruction information to a conveyance management device configured to manage the plurality of automatic guided conveying apparatuses each conveying the sheet stacking apparatus.

12. A non-transitory computer readable storage medium storing a computer program that, when executed by a processor, causes the processor to:

generate a conveyance instruction information for conveying, by one of a plurality of automatic guided conveying apparatuses, a sheet stacking apparatus in which a sheet ejected from a printer is stacked to a sheet supply position of a next processing machine, based on a job information in which a manufacturing process procedure for manufacturing a printed product is registered, wherein the conveyance instruction information includes an information concerning an orientation of the sheet to be ejected from the printer to the sheet stacking apparatus; and

transmit the conveyance instruction information to a conveyance management device configured to manage the plurality of automatic guided conveying apparatuses each conveying the sheet stacking apparatus.

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